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Volume 03 — High-Speed Links, Wireless, and VPN

Course: CEC / SWAYAM Information Security · **Coordinator:** Dr. Maninder Singh, Thapar Institute

Modules M24-M35: Ethernet security, Gigabit/10-Gigabit Ethernet, ISDN, SCTP, ATM, wireless LANs, Wi-Fi security, Bluetooth, VPN, and WiMAX.

These notes are grounded in the official YouTube lecture transcripts. They are study material, not an official CEC publication.

M24: Ethernet Security

Source: <https://www.youtube.com/watch?v=uaDXhq0k5sc> **Instructor / expert:** Prof. Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar. Course coordinator: Dr. Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala.

Learning objectives

- Trace Ethernet from ALOHAnet and Xerox CSMA/CD through the DIX 1.0 spec to IEEE 802.3, and name the frame as the on-wire unit.
- State why Ethernet was designed without security, and list the five attack classes the lecture uses (access, confidentiality, integrity, DoS, system security).
- Contrast router-based segmentation, access control (including IEEE 802.1Q / 802.1X), higher-layer crypto, and monitoring.
- Describe IEEE 802.1AE MACsec: SecTAG and ICV placement, SA/SAK/PN/SCI, Protect/Validate, and replay protection.

Core concepts

Ethernet security is the lecture's topic for **securing wired LANs and MANs**. The expert's outline is: brief Ethernet background, **Ethernet threats**, **Ethernet security solutions**, then the **MACsec** protocol.

Roots, DIX, and IEEE 802.3 Ethernet's roots are in the 1970s radio experiment **ALOHAnet**, developed under **Norman Abramson** at the **University of Hawaii**. Around the same time **Xerox** created a LAN at **3 Mbps** using **carrier-sense multiple access with collision detection (CSMA/CD)**.

In **1980**, the **10 Mbps Ethernet Version 1.0** specification was jointly released by **Digital Equipment Corporation**, **Intel**, and **Xerox** (the DIX spec). IEEE picked it up as **IEEE 802.3** in **1983**.

A chunk of data Ethernet sends on the wire is a **frame**. Only one node should transmit a frame at a time; simultaneous transmissions cause a **collision**, both signals can fail, and the stations must **retransmit**. Those properties are summarized as **CSMA/CD**.

Ethernet supports **optical fiber** and **copper**. In the lecture figure, yellow is fiber and blue is **twisted pair**; a server farm and PCs attach to a **Fast Ethernet (100 Mbps)** network through **switches**.

IEEE 802.3 comprises several **OSI physical-layer** wiring and signaling variants. Original **10BASE5** used **coaxial cable** as a shared medium; newer variants use **twisted pair** and **fiber** with **hubs or switches**.

Name the lecture uses	Rate
Ethernet	10 Mbps
Fast Ethernet	100 Mbps
Gigabit Ethernet	1,000 Mbps
10 Gigabit Ethernet	10,000 Mbps
100 Gigabit Ethernet	100,000 Mbps
400 Gigabit Ethernet	expected by early 2017

Rates have risen from the original **10 Mbps** to **100 Gbps**, with **400 Gbps** expected by early 2017 in the lecture's timeline. Ethernet is called the **only survivor of the wired**

LAN war: it is hard to find an IP packet that has not crossed an Ethernet segment, largely because of **simplicity and ease of configuration**. It has nonetheless always been known as an **insecure** technology. Successful **malware** and the move toward **cloud computing** in data centers are why security of Ethernet now needs attention.

Why Ethernet is easy to attack Security was **never a major design consideration**. The basis of an attack is **gaining access to the target Ethernet segment**. The attacker may be an **insider with full access rights**, or may have found a jack in a **public space**.

Before listing methods, the lecture lists how an attacker **uses** that access:

- Learn **private network topology** and **traffic** for later attacks
- Gain **control of switches, routers, or servers** in the LAN
- **Eavesdrop**
- **Manipulate** information
- **Disrupt availability**

The **most prominent** attack methods on Ethernet segments are:

1. **Network and system access**
2. **Traffic confidentiality**
3. **Traffic integrity**
4. **Denial of service**
5. **System security** (named as a class; the worked examples sit under the first four)

Network and system access Access is a **prerequisite** for every other attack. It is achieved by **connecting equipment** or by **taking over existing resources**.

Method	What the lecture describes
Unauthorized join	Connect to an unused switch port : physical access to the switch, a wall socket , unplug a PC and plug in another, or insert a switch between PC and wall jack
Unauthorized expansion	Users install their own switches or wireless access points , so others can join
VLAN join	If a switch listens for VLAN management protocols on host ports , a host can act as a switch and join all VLANs . Some ports can be set not to transmit those protocols but still listen ; an attacker can probe for hidden features
VLAN tagging and hopping	Craft Ethernet frames with a VLAN tag and inject into VLANs the attacker should not reach
Remote access to the LAN	Compromise a host via a higher layer , e.g. social engineering so a user opens a remote administration service that connects out to the Internet , giving the attacker an Ethernet-layer foothold

Method	What the lecture describes
Switch control	Switches often ship with default or no passwords ; passwords can often be physically reset . A controlled switch can reroute traffic , take links down , claim the STP root by raising bridge priority , or DoS selected links

Traffic confidentiality Captured traffic is useful in itself and for **finding targets**. The attacker obtains not only payload but **authentication data such as passwords** and **topology** for later use.

Traffic integrity The attacker **modifies** traffic. Example: **imitate a bank web server to the user** and **imitate the user to the bank**, gaining **temporary control of the user's bank account**.

Technique	Mechanism
ARP poisoning	Send an ARP reply with the victim's IP and the attacker's MAC , so the sender delivers frames to the attacker
DHCP poisoning	Hear broadcast DHCP requests and reply first ; on success assign gateway , DNS , and IP , then control the host's traffic
Man-in-the-middle	If traffic is steered through the attacker and has no integrity check , the attacker can modify it
Session hijacking	Ethernet is stateless , but many higher-layer protocols create a session that is then assumed trusted with no further verification . If the attacker eavesdrops (or otherwise learns enough), they can recreate the session and act as one endpoint

Denial of service Motivation is **not data access** but **preventing use: total loss** or **degradation** of service.

- **Resource-exhaustion:** target a switch's **control and management planes** with frames that need **extra processing**.
- **Protocol-based DoS: Spanning Tree Protocol (STP)** builds a **tree from a mesh** and is **self-configuring**. A node can send **STP messages** and **pretend to be a switch**. The switching network can be **halted** by flooding **STP topology-change notifications (TCNs)** or other STP control messages.

Solutions the lecture groups Ethernet security has been improved by **standards bodies, vendors, and research**. The traditional answer was to treat **any Ethernet**

segment as untrusted and put it **inside a protected domain: behind a firewall**, in a **secure building**, with **trusted staff**. Remaining issues were left to **higher-layer cryptography** such as **IPsec** and **TLS**. Cryptography has **its own cost** and is **not a universal solution**.

Router-based security Replace a **central Ethernet switch** with an **IP router**. The router **partitions** Ethernet into **several segments**, each a **separate broadcast domain**. **ARP, STP, VLAN, and MAC-address-table** attacks then **cannot cross segments**. Inside a segment the same attacks remain unless **each switch** is replaced by a **multi-port router**. Cost: an IP router needs **configuration** (address allocation, default route); that can sometimes be **automated** (e.g. **residential broadband** where topology is clear to the access router). Routers also **hinder easy mobility**: on Ethernet a host can **move and keep IP and MAC**; **MAC tables update automatically**. Compared with a switch, a router still gives **considerable protection** against other users on the same box.

Access control If untrusted parties never get on, they cannot attack.

- **Physical protection:** equipment in **locked cabinets and rooms**; wiring **inside walls**.
- **Segmentation and VLANs:** **IEEE 802.1Q** limits **broadcast and other traffic** to specific segments. VLANs are **logically separate** on the same physical plant and define **security domains**. Each host can be put in its **own VLAN**. **IEEE 802.1Q-in-Q (QinQ) double tagging** or vendor **private VLAN (PVLAN)** isolates hosts so traffic only reaches one **promiscuous port** (typically a **router to the Internet**). Each host sees **only itself and the promiscuous-port host**; only a **few VLAN IDs** are needed on trunks for PVLAN traffic.
- **Access control lists:** **not part of Ethernet** itself; **switch vendors** added them. A simple Ethernet-frame ACL can use **sender/receiver MAC** or **EtherType**. Access can be limited by **MAC**, and **service-specific ACLs** are common.
- **Authentication-based access:** **IEEE 802.1X** port authentication supports **username/password** or a **certificate and private key**. The **client** talks to the **authenticator switch**; the switch checks credentials against an **authentication-server database**. 802.1X uses **Extensible Authentication Protocol (EAP)** for many methods. It authenticates a host at **session start** and **binds the MAC address to a switch port**.

Secure protocols on the wire

- **Encryption and integrity:** **IEEE 802.1AE MACsec** builds **encrypted connections between hosts and switches**, protecting **confidentiality and integrity of frame contents**. Deployment needs **software** and **authentication configuration** for each participating entity. MACsec stops **intruders** from **reading or modifying** data frames (detail below).
- **Securing ARP:** ARP is a **major Ethernet vulnerability**. **DHCP snooping** can bind **MAC ↔ IP ↔ port** from DHCP messages and **block ARP spoofing**. Limitation: a **single switch cannot see** DHCP allocations whose path to the server **does not pass through that switch**.

Security monitoring The previous techniques are mostly **proactive** and, once set, need little **external or human** involvement. **Active** monitoring adds:

- **Ethernet firewall and deep packet inspection (DPI):** firewalls sit between segments and are **more complex ACLs**, including **stateful** features; current products can operate on **all network layers**.
- **IDS / IPS:** use DPI against a **signature library** of known attacks. The lecture states such systems are **rarely useful for Ethernet-network protection**.
- **Planning, configuration, and administration:** good practice **strongly influences** Ethernet security; many of the technical controls need **configuration and ongoing adjustment** as topology changes.

IEEE 802.1AE MACsec **MAC Security (MACsec)**, IEEE 802.1AE, lets **authorized systems** that attach to and interconnect LANs **keep transmitted data confidential** and **resist frames sent or modified by unauthorized devices**.

The lecture's list of what MACsec facilitates:

- Maintain **correct network connectivity and services**
- **Isolate denial-of-service** attacks
- **Localize** any source of communication to the **LAN of origin**
- Build **public networks** that serve **unrelated or mutually suspicious customers** on **shared LAN infrastructure**
- **Secure communication between organizations** that use a LAN for transit
- **Incremental, non-disruptive** deployment, protecting the **most vulnerable** components first

On the frame: destination and source **MAC addresses stay as they are**. A **SecTAG** is inserted **before** the user data; an **integrity check value (ICV)** is inserted **after** the data. MACsec **modifies the MAC service data unit (MSDU)** carried in each frame.

The **MAC security tag (SecTAG)** carries parameters that **identify the protocol**, **identify the key** used to validate the frame, and support **replay protection**:

SecTAG parameter	Role
Tag Control Information (TCI)	Tag/control bits
Association Number (AN)	Selects the Security Association
Packet Number (PN)	Per-SA packet counter (replay)
Secure Channel Identifier (SCI)	Identifies the secure channel

The **secure data** field carries user data, **encrypted if confidentiality is provided**. The **ICV** covers **MAC DA, MAC SA, SecTAG, and user data**.

Transmit path

1. The frame is assigned to a **Security Association (SA)** identified locally by its **Association Number (AN)**.
2. The SA identifies the **Security Association Key (SAK)** and the **next Packet Number (PN)**.

3. **AN, SCI, and PN** are encoded in the SecTAG and given to the **Protect** function.
4. Protect emits the **ICV** and **secure user data**.

Receive path

1. **AN, SCI, and PN** are extracted from the SecTAG.
2. **AN** and **SCI** assign the frame to an SA and thus to the **SAK**.
3. The current **cipher suite's validation** function is given **SAK, PN, SCI, DA, SA, SecTAG octets, secure data, and ICV**.
4. If integrity holds and user data can be decoded, a **valid** indication and the **user-data octets** are returned.
5. **Replay protection:** the received **PN must not be less than the lowest acceptable PN** for that SA.
6. On success, unchanged frame parameters are presented to the MACsec receiver and the **lowest acceptable PN is updated**.

Key terms

Term	Meaning in this lecture
Frame	Ethernet's on-wire data unit; only one node should send one at a time
CSMA/CD	Carrier sense, multiple access, collision detection — Ethernet's classic shared-medium rule
IEEE 802.3	LAN/MAN standard grown from DIX Ethernet 1.0 (1980) and standardized in 1983
VLAN hopping	Injecting VLAN-tagged frames (or abusing VLAN management) to reach VLANs the port should not join
STP TCN flood	DoS by flooding Spanning Tree topology-change notifications or other STP control
IEEE 802.1Q / QinQ / PVLAN	VLAN tagging, double tagging, and private-VLAN isolation (hosts only talk via a promiscuous router port)
IEEE 802.1X / EAP	Port authentication of a host to a switch that checks an authentication server
DHCP snooping	Bind MAC-IP-port from DHCP so ARP spoofing can be blocked, limited to what that switch can see
MACsec (IEEE 802.1AE)	Hop-by-hop encryption/integrity of Ethernet MSDUs using SecTAG, SAK, PN, SCI, and ICV
SecTAG / ICV / SAK / PN	MACsec tag, integrity check, association key, and packet number used for Protect, Validate, and replay checks

Lecture takeaways

- Ethernet won wired LANs by simplicity, not by security; design never treated security as a first-class goal.
- Almost every attack starts with **segment access** (physical jack, rogue AP/switch, VLAN tricks, stolen switch, or a remotely owned host).
- Confidentiality, integrity (ARP/DHCP poisoning, MITM, session hijack), and STP/resource DoS are the concrete Ethernet threats taught here.

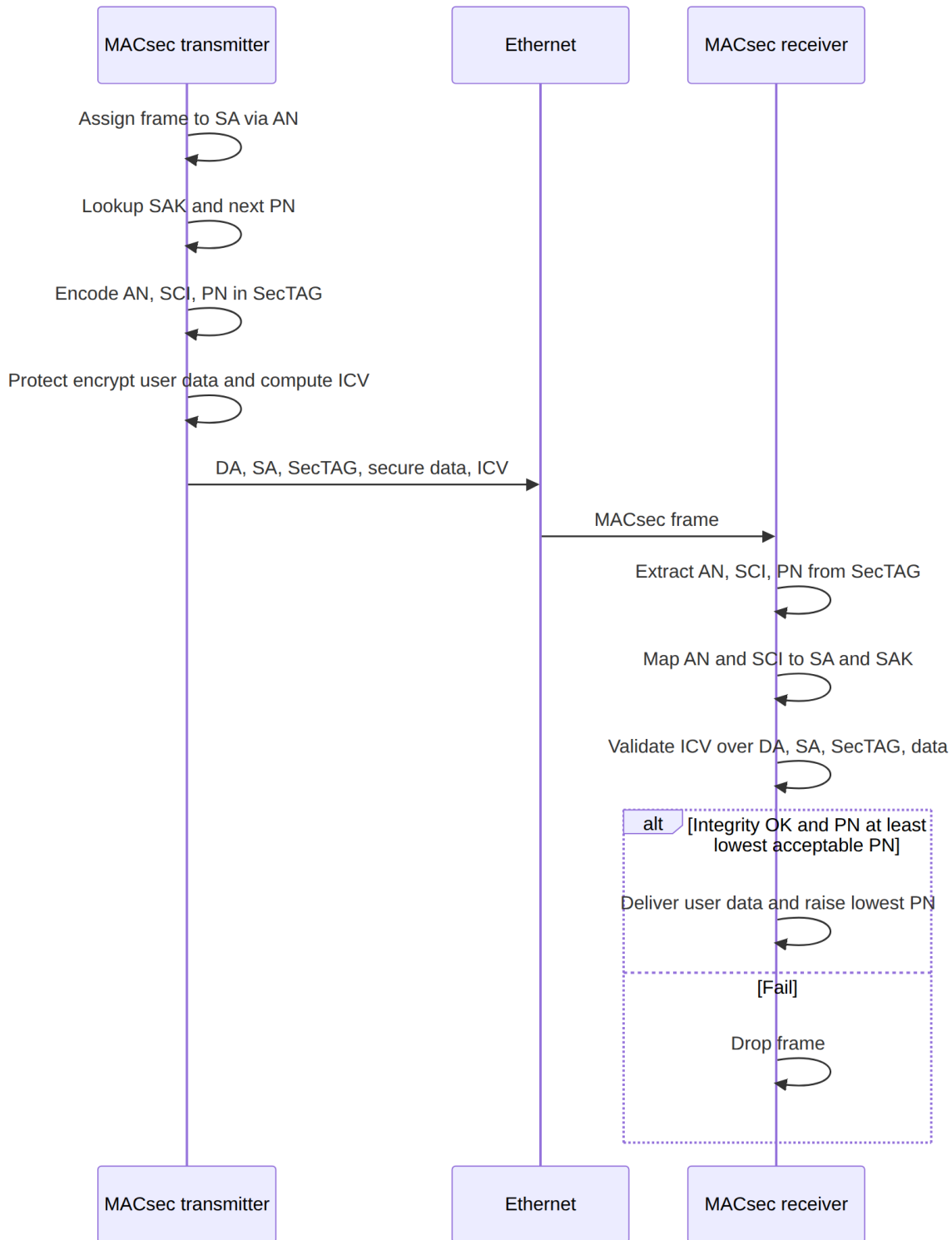


Figure 1: Architecture diagram

- Classic mitigation was “treat the LAN as dirty” (building + firewall) plus **IPsec/TLS**; that is costly and incomplete.
 - Routers cut broadcast-domain attacks; 802.1Q/PVLAN/ACLs/802.1X limit who is on the wire; DHCP snooping hardens ARP; firewalls/IDS are secondary.
 - **IEEE 802.1AE MACsec** is the Ethernet-layer protocol the lecture spends its last section on: encrypt and integrity-protect frames between authorized devices, with replay protection via PN.
-

M25: Gigabit Ethernet

Source: <https://www.youtube.com/watch?v=4rAKMS0KbJU> **Instructor / expert:** Dr. Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar. Course coordinator: Dr. Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala.

Learning objectives

- Explain why Gigabit Ethernet (1 Gbps / 1000 Mbps) was specified as a 10× Fast Ethernet successor that keeps the IEEE 802.3 frame and MAC modes.
- Map the Gigabit stack: GMII, PCS (8B/10B), PMA, PMD/MDI, and auto-negotiation of 10/100/1000 Mbps and half/full duplex.
- List the IEEE 802.3z / 802.3ab / 802.3ah (and related) physical types with media, wavelengths, and distances the lecture gives.
- Recite the basic Gigabit MAC frame fields, CSMA/CD vs full duplex, packet bursting, and **carrier extension**.

Core concepts

Gigabit Ethernet is presented as **wired-LAN transmission at 1000 Mbps (1 Gbps) — ten times Fast Ethernet**. The lecture order is: introduction, **layered protocol architecture**, **physical specifications**, then **frame format**.

Why 1 Gbps became necessary A “basic law of network design” in the lecture: **demand for capacity is always underestimated**. Gigabit speeds once looked excessive; **data-intensive applications**, more users, and new delivery methods keep raising bandwidth need.

When **100 Mbps** technologies such as **FDDI** appeared, most **horizontal** networks still used **10 Mbps Ethernet**; the new protocols were mainly **backbones**. Once **Fast Ethernet** took the horizontal market, a **100 Mbps backbone** was often **too small** for **switch-to-switch** links that aggregate many Fast Ethernet networks. Gigabit Ethernet was developed as the **next generation** at **1 Gbps**.

IEEE standards the lecture names

Standard	Year (lecture)	What it defined
IEEE 802.3z	June 1998	Initial Gigabit Ethernet; required optical fiber . Commonly 1000BASE-X , where X is CX, SX, LX , or non-standard ZX
IEEE 802.3ab	1999	Gigabit over UTP Cat 5 / 5e / 6 , known as 1000BASE-T . After ratification, Gigabit became a desktop technology on existing copper
IEEE 802.3ah	2004	Two more fiber PHYs: 1000BASE-LX10 and 1000BASE-BX10 , in the Ethernet in the First Mile group

Like Fast Ethernet, Gigabit uses the **same frame format, frame size, and media-access method** as **10 Mbps Ethernet**. Fast Ethernet overtook FDDI because administrators did not need a **different backbone protocol**; likewise Gigabit avoids forcing **ATM** on backbones.

It is an **extension of base IEEE 802.3**. Task-force **design objectives**:

- Offer **10× the bandwidth of Fast Ethernet**
- Use the **IEEE 802.3 Ethernet frame format**
- Employ the same **half-duplex and full-duplex MAC** schemes as predecessors
- Be **backward compatible** with **10 Mbps and 100 Mbps Ethernet**
- Support **all existing network protocols** used with the Ethernet family

Features emphasized

- Same **802.3 frame size and format** → easy integration
- **Upgrade path** that reuses existing technology and operational knowledge
- **Full duplex** for **switch-switch** and **switch-end-station** links; **most products shipped are full duplex**, so **no shared-medium contention**
- **Half duplex** on shared media still uses **CSMA/CD**; a **packet-bursting** feature lets servers, switches, and other devices **send bursts of small packets** to use the bandwidth
- Media: **fiber**, or **Cat 5 / Cat 6**
- **Low cost** of install, maintenance, and management (local administrators can run it)
- **Faster switching** than routing: same frame format allows **seamless LAN / MAN / WAN** integration with **no fragmentation, reassembly, or address translation**, so **routers (slower than switches) are not required** for that stitching

Layered protocol architecture Gigabit Ethernet supports **10, 100, and 1000 Mbps**. It provides separate **8-bit-wide receive and transmit data paths**, so both **full duplex and half duplex** are possible.

The **Gigabit Media Independent Interface (GMII)** sits between **MAC and PHY**. It is an extension of Fast Ethernet's **Media Independent Interface (MII)** and **reuses MII's management interface**. With GMII, **shielded/unshielded twisted pair and single-mode/multimode fiber** can share the **same MAC controller**. GMII also provides two media-status signals: **carrier present** and **collision absent**.

GMII sits above three sublayers:

Sublayer	Role in this lecture
PCS (Physical Coding Sublayer)	Uniform interface to all media. 8B/10B coding as in Fibre Channel: 10-bit code groups represent 8-bit groups. Generates carrier sense and collision detect for half duplex. Runs auto-negotiation : NIC learns 10 / 100 / 1000 Mbps and half vs full duplex
PMA (Physical Medium Attachment)	Medium-independent serial attachment. Takes 10-bit groups at 125 MHz from PCS, serializes them; on receive, deserializes bits back into code groups for PCS
PMD (Physical Medium Dependent)	Maps the medium; defines signaling . Includes the Medium Dependent Interface (MDI) — actual connectors . PMD is where 802.3z, 802.3ab , and related PHYs are distinguished

Physical specifications The PHY describes **media, electrical/optical properties, and signal interpretation**.

IEEE 802.3z — collectively 1000BASE-X 1000BASE-SX (short wavelength fiber)

- Multimode fiber, **850 nm** near-infrared
- Maximum **220 m** on **62.5/125 μm** fiber with good terminators (the lecture also states it **usually works farther**)
- Modern **50/125 μm** fiber can reliably reach **500 m or more**
- Popular for **intra-building** links in large offices, **colocation** facilities, and **carrier-neutral Internet exchanges**

1000BASE-LX (long wavelength)

- Long-wavelength laser, **1270-1355 nm**; typically specified **1300 or 1310 nm**
- Specified to **5 km** over **10 μm single-mode fiber**; often works **much farther**; many vendors guarantee **10 or 20 km** if their equipment is at **both ends**
- For links **greater than 300 m**, a **launch-conditioning patch cord** may be required so the laser is launched at a **precise offset from the core center** and **spreads across the core diameter**

1000BASE-CX

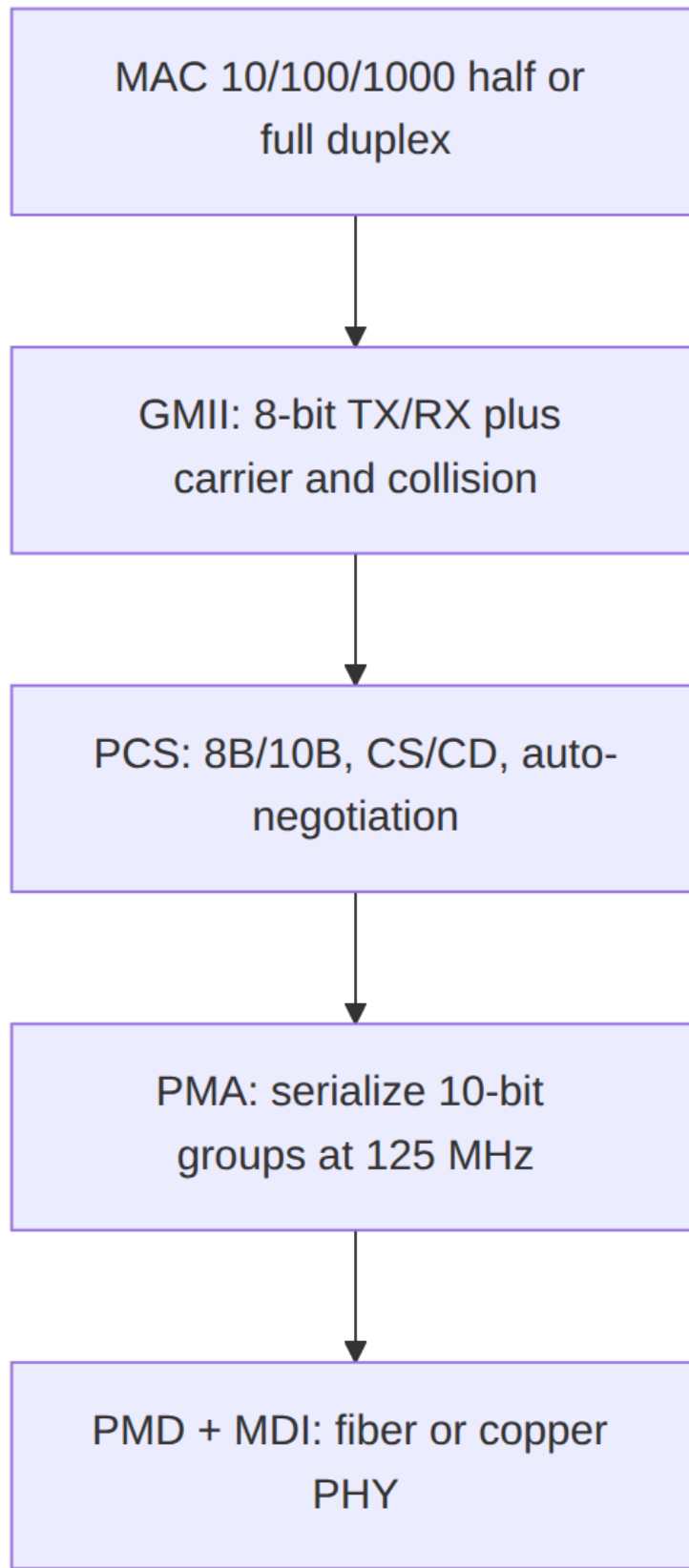


Figure 2: Architecture diagram

- Early copper Gigabit, **maximum 25 m, balanced shielded twisted pair, pinout different from 1000BASE-T**
- Short reach because of the **very high signal rate**
- Still used in niches such as **blade server to switch-module Ethernet**; **1000BASE-T succeeded it** for general copper wiring

IEEE 802.3ab — 1000BASE-T (and the TIA's 1000BASE-TX) **1000BASE-T** is Gigabit on UTP. The **Telecommunications Industry Association (TIA)** promoted a simpler variant, **1000BASE-TX**, meant to cut electronics cost by using **only two pairs in each direction**. Many **1000BASE-T** products are **advertised as 1000BASE-TX** from ignorance that **TX is a different standard**.

Backplane and Ethernet in the First Mile PHYs The lecture groups these in a table (it names **IEEE 802.3ap** in one breath with EFM; IEEE names: **802.3ap** is backplane, **802.3ah** is Ethernet in the First Mile):

1000BASE-KX — part of **IEEE 802.3ap**, Ethernet over **electrical backplanes**. Defines **one to four lanes** of backplane links: **one RX and one TX differential pair per lane**, at bandwidths from **megabit to 10 gigabit per second**.

1000BASE-LX10 — standardized **six years after** the first Gigabit fiber PHYs in the **Ethernet in the First Mile** task group. Very similar to **1000BASE-LX** but **up to 10 km** over a **pair of 1310 nm single-mode fibers**, thanks to **higher-quality optics**.

1000BASE-BX10 — **up to 10 km** on a **single strand of SMF, different wavelength each way**. Ends are **not interchangeable**: **downstream** (network center → outside) uses **1490 nm**; **upstream** uses **1310 nm**.

Non-standard but industry terms **1000BASE-EX** — non-standard but accepted term; similar to LX10 with **up to 40 km** over a **pair of SMF, 1310 nm**, higher-quality optics than LX10.

1000BASE-ZX — multi-vendor term: **1550 nm, at least 70 km SMF**. Some vendors specify **up to 120 km**, sometimes called **1000BASE-EZX**.

Hardware to upgrade 10/100 networks Four hardware types:

1. Gigabit Ethernet **NICs**
2. **Aggregating switches** that connect many Fast Ethernet segments **into** Gigabit Ethernet
3. **Gigabit Ethernet switches**
4. **Gigabit Ethernet repeaters**

CSMA/CD, duplex, and the frame The Gigabit **MAC** uses the same **CSMA/CD** protocol as Ethernet. **Maximum cable-segment length** is limited by CSMA/CD: if two stations see idle media and transmit, a **collision** occurs.

IEEE 802.3z MAC operation is **half or full duplex**.

- **Half duplex:** can send or receive, **not both at once**; uses classic **CSMA/CD**.
- **Full duplex:** send and receive **at the same time**. The Gigabit MAC then uses **IEEE 802.3x** full duplex, including **IEEE 802.3x flow control**. Full duplex raises point-to-point bandwidth from **1 to 2 Gbps**, **increases maximum distance** for a given medium, and **does not use CSMA/CD** because **collisions are eliminated**. Full duplex is **best for backbones** and **high-speed servers**.

An enhancement from **switched 100 Mbps Ethernet** is kept: in half duplex, **CSMA/CD** remains, with **packet bursting** of small frames.

Basic data frame (seven fields; optional formats exist) The lecture's **required** basic format:

Field	Size	Meaning
Preamble (PR)	7 bytes	Alternating 1s and 0s: "a frame is coming," and PHY receive synchronization
Start of Frame (SOF / SFD)	1 byte	Alternating 1s and 0s ending in two consecutive 1s ; next bit is the leftmost bit of the leftmost byte of DA
Destination Address (DA)	6 bytes	Who should receive. Leftmost bit: 0 = individual, 1 = group. Second bit: 0 = globally administered, 1 = locally administered. Remaining 46 bits uniquely identify a station, a group, or all stations
Source Address (SA)	6 bytes	Sender; always individual; leftmost bit always 0
Length/Type	2 bytes	If value ≤ 1500 , it is the count of LLC bytes in Data. If > 1500 , the frame is an optional type and the field is a type ID
Data	$n \leq 1500$ bytes	If shorter than 46 bytes , pad so Data is 46 bytes
Frame Check Sequence (FCS)	4 bytes	32-bit CRC computed by the sending MAC over DA, SA, Length/Type, and Data ; receiver recalculates to detect damage

Carrier extension Gigabit Ethernet must stay **interoperable with existing 802.3** networks. **Carrier extension** keeps **IEEE 802.3 minimum and maximum frame sizes** while allowing **meaningful cable distances**. For a carrier-extended frame, **extension symbols are included in the collision window** — the **entire extended frame** is considered for collision and dropped if it collides — but **FCS is calculated**

only on the original frame. The receiver **strips extension symbols before checking FCS.**

Key terms

Term	Meaning in this lecture
Gigabit Ethernet	1000 Mbps / 1 Gbps IEEE 802.3 family, 10× Fast Ethernet
1000BASE-X	IEEE 802.3z fiber/short-copper group (SX, LX, CX; ZX as vendor term)
1000BASE-T	IEEE 802.3ab Gigabit on Cat 5/5e/6 UTP
GMII	Gigabit Media Independent Interface between MAC and PHY
8B/10B	PCS coding: 8 data bits as 10-bit code groups (Fibre Channel style), 125 MHz toward PMA
Packet bursting	Half-duplex CSMA/CD enhancement: burst small packets to fill the pipe
IEEE 802.3x	Full-duplex operation and flow control used by Gigabit when not sharing the wire
Carrier extension	Extra symbols in the collision window so min/max 802.3 sizes still work at 1 Gbps distances; FCS ignores the extension
Length/Type	≤1500 → LLC length; >1500 → optional Ethernet type

Lecture takeaways

- Gigabit exists because Fast Ethernet desktops overflowed 100 Mbps backbones; it keeps **802.3 frames** so shops need not move backbones to **ATM** or **FDDI**.
- Architecture is **MAC → GMII → PCS (8B/10B, auto-neg) → PMA (125 MHz serialize) → PMD**.
- Fiber PHYs (SX/LX/LX10/BX10/EX/ZX) and copper (CX then T) are chosen by **reach and cabling already in the wall**.
- **Full duplex** is the shipping default (up to **2 Gbps** both ways, no CSMA/CD); half duplex still exists with **CSMA/CD + burst + carrier extension**.
- Frame layout is classic Ethernet (preamble through 32-bit CRC); **carrier extension is the Gigabit-specific MAC trick** taught at the end.

M26: 10 Gigabit Ethernet

Source: <https://www.youtube.com/watch?v=yu-UWXqj5wE> **Instructor / expert:** Dr. Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar. Course coordinator: Dr. Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala.

Learning objectives

- State that 10 Gigabit Ethernet (10 GbE / 10G / 10 GigE) sends Ethernet frames at **10 Gbps**, first in **IEEE 802.3ae-2002**, and that it is **full duplex only** (no hubs, no CSMA/CD).
- Walk the 10G stack: RS, PCS, PMA, PMD, XGMII, MDI, and the **WAN Interface Sublayer (WIS)** for SONET-friendly WAN PHYs.
- Contrast **serial 10 Gbps** vs **parallel n-lane** (including WDM) implementations.
- Recite the unchanged Ethernet MAC frame, why the minimum is **64 octets** without carrier extension, and the LAN-fiber / WAN-fiber / copper / backplane / EPON PHYs with rates, wavelengths, and reaches.

Core concepts

10 Gigabit Ethernet is **wired-LAN (and WAN) transmission at 10 Gbps**. Outline: introduction, **layered architecture**, **PHY and MAC**, then **10G standards**.

Demand, HSSG, and what 10G is As high-speed demand grew, a faster Ethernet was needed. In **March 1999** the **Higher Speed Study Group (HSSG)** formed to develop a **10 Gigabit Ethernet** standard. 10 GbE is described as a **telecommunication technology** offering up to **10 billion bits per second**.

10 Gigabit Ethernet (also **10G**, **10GbE**, **10 GigE**) is a group of technologies that transmit **Ethernet frames at 10 Gbps**. It was first defined by **IEEE 802.3ae-2002**.

Unlike earlier Ethernet, **10 GbE defines only full-duplex point-to-point links**, generally via **network switches**. **Shared-medium CSMA/CD was not carried forward**. **Half duplex and hubs do not exist** in 10GbE.

It can use **copper or fiber**, but bandwidth forces **higher-grade copper**: **Category 6, 6A, or 7** for links **up to 100 m**.

IEEE 802.3 documents listed in the lecture table

IEEE document	Year	Lecture's scope
802.3ae	ratified 2002	10 Gbps over fiber for LAN and WAN
802.3ak	2004	10 Gbps over twinax
802.3-2005	2005	Base-standard revision incorporating prior amendments
802.3an	2006	10 Gbps over copper twisted pair
802.3ap	2007	Backplane Ethernet: 1 and 10 Gbps over printed circuit boards
802.3aq	2006	10 Gbps over multimode fiber with enhanced equalization
802.3-2008	2008	Another base-standard revision
802.3av	2009	10 Gbps Ethernet PHY for EPON

IEEE document	Year	Lecture's scope
802.3-2012	2012	Called the latest version of the base standard in this lecture

Layered protocol architecture Ethernet implements the **bottom two OSI layers: data link and physical**. Mapping in the lecture:

- OSI **physical** → **PMD, PMA, PCS**, and **Reconciliation Sublayer (RS)**
- Toward the **network** side → **MAC** and **LLC**

Basic media families:

- **10GBASE-R** — LAN fiber
- **10GBASE-W** — WAN fiber
- **10GBASE-X** — copper (coded family)

PMD — Physical Medium Dependent. Signaling on the wire/fiber: **amplify, modulate, wave-shape**. Different PMDs support different media.

PCS — Physical Coding Sublayer. Coding plus **serializer / multiplexing**. This structure **distinguishes LAN vs WAN PHYs**:

- **WAN PHY:** PCS operates **serialized**.
- **LAN PHY:** two modes:
 1. **Parallel:** a multiplexer puts data on **four 2.5 Gbps** lines.
 2. **Serial LAN:** serialize/deserialize a **single channel**.

Serial LAN uses **64B/66B** coding. For the **serial WAN PHY**, an extra **WAN Interface Sublayer (WIS)** sits between **serial PCS** and **serial PMA**. WIS **ensures operability with SONET** by including a **simplified SONET frame**.

PMA — Physical Medium Attachment. Serializes **code groups** into a bit stream for serial devices (and the reverse). Supports **multiple encodings** for PMDs whose encoding is **medium-specific**.

Reconciliation Sublayer (RS). **Command translator:** maps **MAC terminology and commands** into **electrical formats** for PHY entities.

MAC. Logical connection between MAC clients and the peer. **Initialize, control, and manage** the peer connection.

XGMII — 10 Gigabit Media Independent Interface. Standard **MAC ↔ PHY** interface; **isolates** the MAC so it can run over **various PHYs**.

MDI — Medium Dependent Interface. **Connector types** for each medium and PMD.

Serial vs parallel PHY implementations Anyone used to Ethernet notices **two physical-layer options**.

Serial: one high-speed **10 Gbps PCS-PMA-PMD** block; **one physical channel at 10 Gbps**.

Transmit: RS passes the MAC data **word by word** to PCS → PCS **encodes** → PMA **serializes** → PMD sends the stream on fiber at **10 Gbps**. Receive is the reverse. **Advantage:** straightforward; **no complicated mux/demux**.

Parallel: n subchannels (parallel cables or **WDM**). A **distributor** multiplexes MAC data into **n streams in round-robin**; each stream goes to its own PCS → PMA → PMD at **(10 / n) Gbps**. Receive uses a **collector**. **Advantage:** lower PCS/PMA rate, so **cheaper CMOS or bipolar** parts. **Disadvantages:** distributor/collector **sensitive to timing jitter**; **multiple logic sets and lasers**.

MAC layer of 10 GbE Very similar to earlier Ethernet MACs: **same addresses and frame formats**, but **no half duplex**. It can support **data rates less than 10 Gbps** using a **pacing mechanism** for **rate adaptation and flow control**. **Full duplex only**.

Why drop half duplex. Original Ethernet's half duplex used **CSMA/CD** on a shared medium; simplicity helped Ethernet succeed, so many people **wrongly treat CSMA/CD as "standard Ethernet."** Half duplex's problems are **efficiency** and **distance**: link distance is limited by **minimum MAC frame size**, which **hurts high-rate** efficiency. At **10 Gbps**, half duplex is **not attractive**; **no realistic market** exists, because most 10 Gbps links are **point-to-point optical**. Full duplex has **no contention**: the MAC may send whenever the **peer is ready to receive**. So the 10G standard specifies **full duplex only**.

Frame format. The point of 10G was to **keep the same MAC frame** as earlier Ethernet for **seamless integration**. With full duplex, **link distance does not affect MAC frame size**. **Minimum MAC frame size is 64 octets**, as in previous standards. **Carrier extension is not required**.

Field	Size	Meaning (same story as Gigabit Ethernet)
Preamble	7 bytes	Alternating 1/0; announces a frame; PHY receive sync
Start of Frame Delimiter (SFD)	1 byte	Alternating 1/0 ending in two 1s ; next bit is leftmost bit of DA
Destination Address	6 bytes	Leftmost bit 0 = individual, 1 = group; second bit 0 = global, 1 = local admin; remaining 46 bits identify station/group/all
Source Address	6 bytes	Sender; always individual; leftmost bit always 0
Length/Type	2 bytes	$\leq 1500 \rightarrow$ LLC byte count in Data; optional formats use type ID
Data	$n \leq 1500$ bytes	If < 46 bytes, pad to 46
FCS	4 bytes	32-bit CRC over DA, SA, Length/Type, Data

(The lecture states $\text{Length/Type} \leq 1500$ as LLC length; it does not repeat the ">

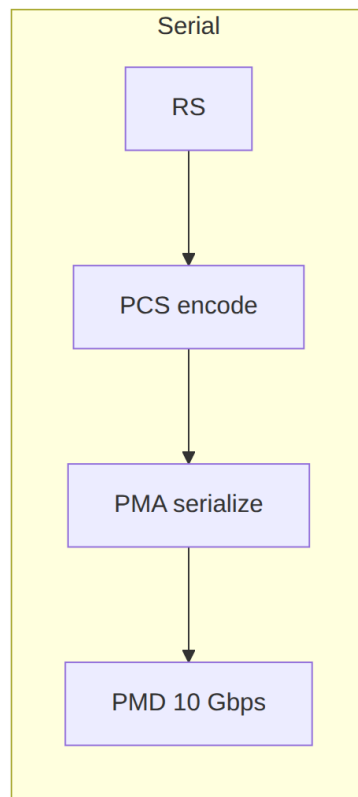
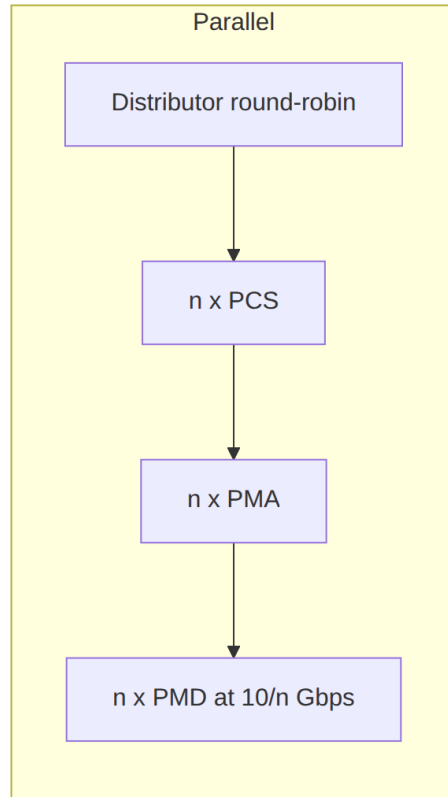


Figure 3: Architecture diagram

1500 means type” sentence in this module the way Module 25 did, but the field is still described as length **or** type ID for optional formats.)

Physical-layer standards, three categories

1. **LAN fiber:** 10GBASE-SR, 10GBASE-LRM, 10GBASE-LR, 10GBASE-ER, 10GBASE-LX4, 10GBASE-ZR
2. **WAN fiber:** 10GBASE-SW, 10GBASE-LW, 10GBASE-EW
3. **LAN copper / backplane:** 10GBASE-CX4, 10GBASE-T, 10GBASE-KX4, 10GBASE-KR

Plus **10GBASE-PR** for PON, taught last.

LAN fiber (“R” PHY) Most common optical variety: **LAN PHY**, used **directly between routers and switches**, letter **R** in the name. Despite the name **LAN**, fiber can run **up to 80 km**. LAN PHY **line rate 10.3 Gbps** with **66-bit** (64B/66B) encoding.

PHY	Year	Medium / λ	Line rate	Reach	Standard
10GBASE002 SR (short reach)		Serial MMF , 850 nm	10.3125 Gbps	300-400 m	802.3ae
10GBASE002 LR (long range)		1310 nm SMF , 64B/66B PCS	10.3125 Gbps	10 km	802.3ae
10GBASE002 ER (ex- tended range)		1550 nm SMF	10.3125 Gbps	40 km ; some vendors later 80 km plug-gables	802.3ae
10GBASE002 LX4		WDM, 1310 nm; four lasers at 3.125 Gbps on unique wavelengths	(4 × 3.125)	300 m on deployed MMF and 10 km SMF	802.3ae; lecture says it is being replaced by 10GBASE-LRM

PHY	Year	Medium / λ	Line rate	Reach	Standard
10GBASE006 LRM (long-reach multi-mode)		1310 nm on FDDI-grade 62.5 μm MMF from early-1990s 100 Mbps plants	10.3125 Gbps	220 m (IEEE 802.3aq figure; ASR in the transcript says “22 m”)	802.3aq

WAN fiber (“W” PHY) 10GBASE-SW, 10GBASE-LW, 10GBASE-EW use the **WAN PHY**. At the physical layer they **correspond to SR, LR, and ER** respectively, so they use the **same fiber types and distances**. They fall under **IEEE 802.3ae**.

Copper LAN and backplane 10GBASE-CX4 — designed 2002; working group **IEEE 802.3ak**. **Four lanes each direction** over copper using **InfiniBand 4X twinax (8-pair)**. **15 m** maximum. **Lowest cost per port** among 10G interconnects, **at the expense of range**. Each copper lane carries **3.125 GHz** of signaling bandwidth.

10GBASE-T — **IEEE 802.3an**, 2006. 10 Gbps on conventional **UTP or STP Cat 6 / 6A / 7**. Cat 6: **55 m**; Cat 6A and 7: **100 m**.

10GBASE-KX4 and **10GBASE-KR** — **backplane Ethernet**, task force **IEEE 802.3ap**, for **blade servers** and **modular routers/switches** with upgradable line cards. Implementations must work with **up to 1 m of copper PCB and two connectors**. Two 10 Gbps port types: **KX4** (four backplane lanes) and **KR** (single lane). **New backplane designs use KR** rather than KX4.

10GBASE-PR (10G EPON / PON) Originally **IEEE 802.3av**. 10G Ethernet PHY for **passive optical networks**. **1577 nm** lasers **downstream**, **1270 nm upstream** (transcript “157 nm” is the IEEE 802.3av downstream wavelength with a dropped digit). Downstream delivers serialized data at **10.3125 Gbps** in a **point-to-multipoint** configuration. Three **power budgets**: **10GBASE-PR10**, **10GBASE-PR20**, and **10GBASE-PR30** (lecture “PR1 / PR20 / PR30”).

Key terms

Term	Meaning in this lecture
10 GbE / 10G / 10 GigE	Ethernet frames at 10 Gbps; full duplex only
IEEE 802.3ae-2002	First 10G Ethernet standard (LAN/WAN fiber)

Term	Meaning in this lecture
10GBASE-R / W / X	LAN fiber, WAN fiber, and copper coding families
XGMII	10 Gigabit Media Independent Interface (MAC-PHY)
WIS	WAN Interface Sublayer: simplified SONET framing between serial PCS and PMA
64B/66B	Serial LAN PCS coding; LAN PHY line rate about 10.3 / 10.3125 Gbps
HSSG	Higher Speed Study Group (March 1999) that started 10G work
10GBASE-T	IEEE 802.3an twisted-pair 10G (Cat 6 55 m; 6A/7 100 m)
10GBASE-PR	IEEE 802.3av 10G EPON PHY with PR10/PR20/PR30 power budgets

Lecture takeaways

- 10 GbE is still Ethernet frames, but **only full-duplex switched point-to-point** — CSMA/CD and hubs are gone because half duplex at 10 Gbps is not a market.
- Stack is classic 802.3 PHY split (**PCS/PMA/PMD**) plus **XGMII** and, for WAN, **WIS/SONET**.
- Implementers choose **one 10G serial pipe** (simple) or **n slower lanes / WDM** (cheaper silicon, more jitter and optics).
- MAC frame is the familiar 64-octet-minimum Ethernet frame; **no carrier extension**.
- PHY letter soup is organized as **LAN-R, WAN-W, copper/backplane, and EPON-PR**, with **10.3125 Gbps** as the repeated serial line rate on optical LAN PHYs.

M27: ISDN

Source: <https://www.youtube.com/watch?v=xWRKLxEw-v4> **Instructor / expert:** Dr. Lal Chand, Department of Computer Engineering, Punjabi University, Patiala. Course coordinator: Dr. Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala.

Learning objectives

- Define ISDN as ITU/CCITT digital circuit- and packet-switched service over ordinary telephone-grade copper (and other media).
- Contrast BRI (2B+D = 144 kbps) with PRI (23B+D / 30B+D) and name B vs D channel rates.
- List bearer, teleservice, and supplementary services, plus TE1/TE2/TA/NT1/NT2 and R-V reference points.
- State ISDN vs ADSL differences, advantages/disadvantages, and the lecture's security conclusion (logical attacks and HTTPS matter more than the physical

medium).

Core concepts

The lecture is titled **ISDN security**. **ISDN (Integrated Services Digital Network)** is a method to transfer **voice and data** with particular data-accessible services. It is a set of **CCITT / ITU-T** standards for **circuit-switched transmission** of data over various media using **ordinary telephone-grade copper wire**. ISDN provides **worldwide digital communication** in the shift toward electronic documents and business transactions. Its **digital nature facilitates adding security**, but it was **deployed with little thought to security**. It offers **digital circuit-switched voice and data** as well as **packet-switched data**.

History: POTS to ISDN Before ISDN, analog **plain old telephone service (POTS)** was the worldwide default. “POTS” originally expanded as **Post Office Telephone Service**; the name stayed after post offices stopped offering telephony. POTS was mainly **copper from the subscriber to the central office**. Limits: **long-distance calls** had to be **routed through operators and switchboards** (unreliable, slow); **static / line noise** disturbed communication.

In the **1960s** the industry began converting analog systems to **digitized packets** and **digital switching**. The UN **CCITT**, now **ITU-T (International Telecommunication Union — Telecommunication Standardization Sector)**, pushed research toward ISDN and **international digitization**, initiated in **1984**.

Two major US networks, **Northern Telecom** and **AT&T**, took first implementation steps but **did not interoperate** with existing telecom equipment and software — a **worldwide setback by 1990**. Then **National ISDN 1 (NI-1)** was made **compatible with existing proprietary equipment**, so users did not have to switch brands or buy new software. NI-1 set procedures for future digital telephony “for everyone.”

ISDN improved **voice quality** and **Internet access** via its **packet-switched** connection. Voice and data ride a **Bearer (B) channel** at **64 kbps** (sometimes **56 kbps**), versus a telephone line the lecture quotes at **52 kbps**. A **Data (D) channel** is used for **controlling network services and signaling** to set up/tear down connections and to carry signaling related to the B channels, at **16 kbps or 64 kbps**.

Future: Broadband ISDN (B-ISDN) — voice, data, and video together on **fiber** at **155 Mbps to 622 Mbps and beyond**; called a major R&D topic.

Types: BRI and PRI Two ISDN access types: **Basic Rate ISDN / Basic Rate Interface (BRI)** and **Primary Rate ISDN / Primary Rate Interface (PRI)**.

Interface	Structure (lecture)	Total
BRI	2 × 64 kbps B + 1 × 16 kbps D	144 kbps — enough for individual users

Interface	Structure (lecture)	Total
PRI (typical North American)	23 × 64 kbps B + 1 × 64 kbps D	1.536 Mbps
PRI (alternate quoted)	30 B + 1 D	1.984 Mbps

BRI is **2B+1D**. The D (Delta) channel is for **link management and signaling**.

PRI **depends on the country**. A typical PRI is **23 B channels at 64 kbps** plus **one 64 kbps D**. One PRI format was **24 DS0s**, looking like a **channelized T1**. Internationally the historical transport rate is **2.048 Mbps (E1; transcript “2.48 Mbps”)**. On E1, **one of 32 DS0s** was already used for control, leaving **31 DS0s** for ISDN, hence **30B+D**.

A “garden variety” **T1** vs PRI: PRI normally uses **23 channels** as **Bearer (B)** for digital voice or data and the **24th** as **Data (D)** for control — **23B+D**. **One D channel can control more than 23 B channels**, so a **24B** ISDN circuit is possible if **control comes from another PRI**.

Features The major feature vs classic telephony: **speech and information on the same line**. ISDN can deliver **voice, data, video, fax** over a **single line**, with **at least two simultaneous connections**. It gives **access to packet-switched networks**. As a **circuit-switched telephone network** it provides **better voice/data quality than analog**. Users can **attach several devices** instead of buying many analog lines. The **B channel** supplies the **greater data rate**.

Three service classes **Bearer services, teleservices, and supplementary services**. Teleservices and supplementary services are **visible to end users**; bearer services are **hidden network parts**.

Bearer services — real-time **digital information** between users; **OSI layers 1-3**. Example: **64 kbps, 8 kHz-structured** speech: 64 kbps plus **8 kHz timing** that structures data into **octet intervals** for **PCM speech**. Because the network **knows the signal is speech**, it may apply **transforms that do not preserve bit integrity** but still yield **good audio**.

Teleservices — higher-level functions on top of bearer service; **OSI layers 4-7**. Examples:

- **Telephony**: speech on a **B channel**, control on **D**
- **Facsimile**: bitmap images on **B**, control on **D**
- **Teletex**: textual/formatted documents on **B**, control on **D**

Supplementary services enhance bearer and teleservices independently:

- **Centrex** — emulates a **private network** with specialized features for a subscriber set
- **Call transfer** — move an active call to a third party
- **Call waiting** — notify a busy user of another incoming call

- **Calling line ID** — calling-party address to the called party

These look like circuit-switched phone features but are **equally applicable to packet-switched data calls**.

How ISDN works on the pair Analog/POTS offers **one transmission channel**, so only **one service at a time** (voice **or** data **or** video). ISDN **logically divides the same pair** into multiple channels:

- **B channel: 64 kbps**. BRI typically has **two B channels** — one often for **voice**, one for **data** — on **one copper pair**.
- **D / Delta channel: line and call setup, 16 kbps** on BRI.

Components, protocols, reference points

Component	Role
TE1 (Terminal Equipment type 1)	Native ISDN: ISDN phone, computer, ISDN fax on the ISDN line
TE2 (Terminal Equipment type 2)	Legacy analog phone, old fax, modem, or other gear that needs a TA
TA (Terminal Adapter)	Lets TE2 (and e.g. Ethernet interfaces) talk to the ISDN network
NT1 (Network Termination type 1)	End of the telco local loop ; start of the customer premises network
NT2 (Network Termination type 2)	Usually absent in homes ; in a large company, the PBX / private telephone system
LT (Line Termination)	Telco physical connection
ET (Exchange Termination)	Telco logical connection from the phones into the phone network

Reference points (letters everyone uses to talk about the parts):

- **R** — between old-style telephone and **TA**
- **S** and **T** — in most homes with **no NT2**, they coincide as **S/T bus**
- **U, V** — with **LT** and **ET**, on the **phone-company** side

Different segments have different **wiring, speeds, and encoding**. ISDN provides **digital transmission over ordinary telephone copper and other media**. It uses **circuit switching** to establish a **physical point-to-point** path from source to destination (the lecture says “permanent” in the sense of an established circuit). ISDN standards from the **ITU** cover **OSI physical, data-link, and network** layers (bottom three).

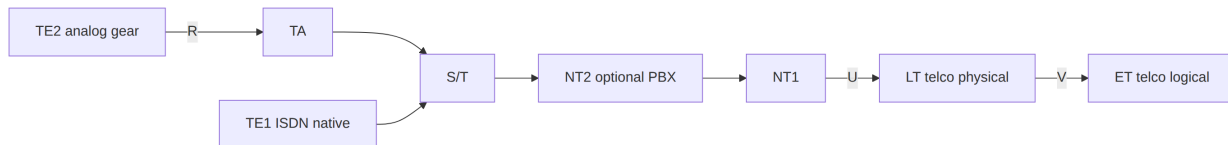


Figure 4: Architecture diagram

ISDN vs ADSL

	ISDN	ADSL (as taught)
Service	Two voice channels or one 128 kbps data channel	Data-only line
Power	Needs local power ; dies if local power fails	Telco copper stays up even when local power fails

Advantages

- **Multiple digital channels concurrently on one copper pair**
- High data rate from the digital scheme: lecture cites **56 kbps**; BRI with **channel aggregation (BONDING or Multilink PPP)** supports **uncompressed 128 kbps** plus overhead/signaling; PRI up to **1920 kbps**
- **Many devices share one line** (faxes, computers, cash registers, credit-card readers); information is **routed to the proper destination**
- **Call setup ~2 seconds** vs **30-60 seconds** for analog modems
- Ringing is not an **in-band ring-voltage** on the B channel: the network sends a **digital packet on a separate channel (out-of-band)**. That **does not disturb** established connections, **takes no B-channel bandwidth**, and makes setup **fast**

Disadvantages

- **More costly** than ordinary telephony / landlines
- Providers and users need **special dedicated Digital Services** (extra cost)

Security issues (ISDN vs cable / DSL / T1) On the **physical medium**, **eavesdropping a POTS landline with dial-up modems is physically easier** than spying on newer media: **lower data rate** suits **homemade electronics**, and some newer protocols include **encryption** that **hinders line tapping**.

The lecture's view: **most attacks now are logical**, not crouching under service boxes at night. Attackers **hack ISP systems or customer machines** (cable/DSL modem or an **unpatched desktop**) by sending **IP packets from far away**. Those attacks are **mostly orthogonal to the medium**. Logical attacks let the attacker try

millions of targets from a basement, instead of weather, cats, and the chance a neighbor calls the police.

Physical spying still exists but is for **specific targets** — the attacker is **after you personally**. Hardening a home network against a **dedicated** attacker is **hard**: think **TEMPEST**, or watching display/keyboard **through a window** with a **telescope** and **high-FPS camera** to record a typed password from about **200 m**.

Advice for DSL, T1, cable, ISDN: choose the ISP with a **good security reputation**, especially for the **modem they provide** — that matters **more than the physical medium**. **Never do sensitive work over the Internet** without **logical protection such as HTTPS**.

Key terms

Term	Meaning in this lecture
ISDN	ITU/CCITT integrated digital voice/data network on phone copper (and other media)
B channel	Bearer channel, typically 64 kbps (sometimes 56), carries user voice/data
D channel	Delta/data channel for signaling and control (16 kbps BRI, 64 kbps PRI)
BRI / 2B+D	Basic Rate Interface, 144 kbps, individual users
PRI / 23B+D / 30B+D	Primary Rate, T1-like 1.536 Mbps or E1-like 1.984 Mbps
B-ISDN	Broadband ISDN on fiber, 155–622 Mbps and beyond
Bearer / teleservice / supplementary	OSI 1–3 network functions; OSI 4–7 user services; add-ons such as Centrex and CLIP
TE1, TE2, TA, NT1, NT2	Native ISDN terminal, legacy terminal, adapter, customer NT, optional PBX
Out-of-band signaling	Setup packets on D, not ring voltage on the user channel
HTTPS	Example of the logical protection the lecture says to use on any medium

Lecture takeaways

- ISDN digitized the local loop: **B channels for payload, D for signaling**, BRI for homes, PRI for sites that need T1/E1-scale bundles.
- Digital operation **could** have made security easier, but ISDN was **rolled out with little security design**.
- Operational wins: shared pair, ~2 s setup, out-of-band signaling, device sharing; costs and special telco service are the downsides.
- Versus ADSL: ISDN can carry voice; ADSL is data-only and often **survives local power loss**.

- Security punchline: **don't fetishize the medium.** Pick a **reputable ISP/modem**, assume **logical IP attacks**, and protect sensitive use with **HTTPS** (and similar).

M28: Stream Control Transmission Protocol (SCTP)

Source: <https://www.youtube.com/watch?v=a277VGgcYi0> **Instructor / expert:** Dr. Sunil Kumar, Department of Electrical and Computer Engineering, Punjab Agricultural University, Ludhiana. Course coordinator: Dr. Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala.

Learning objectives

- Place SCTP as a **reliable, message-oriented** transport between UDP and TCP, aimed at telephony signaling over IP.
- Explain **multistreaming** (association vs TCP's one stream) and **multihoming** (multiple IP addresses, failover not load-share in current implementations).
- Number data with **TSN, SI**, and **SSN**; contrast SCTP **packets/chunks** with TCP **segments**.
- Sketch the **12-byte general header**, chunk layout, and the **four-way handshake** (INIT, INIT ACK, COOKIE ECHO, COOKIE ACK).

Core concepts

Stream Control Transmission Protocol (SCTP) is a **new reliable, message-oriented transport-layer protocol**. In the Internet suite it sits **between application and network**, like TCP and UDP.

It was designed for **recent Internet applications** that need **more sophisticated service than TCP**:

Application (lecture)	Role
IUA	ISDN over IP
M2UA, M3UA	Telephony signaling
H.248	Media Gateway Control
H.323	IP telephony
SIP	IP telephony

SCTP is said to provide the **enhanced performance and reliability** those apps need.

UDP vs TCP vs SCTP **UDP** is **message-oriented**: a process hands UDP a message, which is encapsulated in a **user datagram**. **Message boundaries are conserved**; messages are independent — desirable for **IP telephony** and **real-time data**. UDP is **unreliable**: the sender cannot know fate of messages; they can be **lost, duplicated, or reordered**. UDP also **lacks congestion control and flow control**.

TCP is byte-oriented: messages become a **byte stream** sent in **segments**; **no message boundaries**. TCP is **reliable**: duplicates detected, lost segments **resent**, bytes delivered **in order**, with **congestion and flow control**.

SCTP combines the best of both: reliable and message-oriented. It **preserves message boundaries** and still detects **lost, duplicate, and out-of-order** data, with **congestion and flow control**, plus **innovative features** neither UDP nor TCP has.

Services offered to applications Process-to-process communication. SCTP uses **well-known ports in the TCP port space**. Extra ports listed:

Protocol	Port(s) in the lecture
IUA	9901 (transcript “90”; IETF IUA well-known port)
M2UA	2904
M3UA	2905
H.248	2949
H.323	1718, 1719, 1720 (transcript also garbles “11720”)
SIP	5060

Multiple streams. A TCP connection is **one stream**; a **loss anywhere blocks the rest** — acceptable for text, bad for **real-time audio/video**. SCTP allows **multi-stream service** in each connection, called an **association**. If one stream blocks, **others still deliver**. Analogy: **highway lanes** (regular traffic vs carpool).

Multihoming. TCP uses **one source IP and one destination IP** even if a host is multihomed. An SCTP association lets each end **define multiple IP addresses**. If **one path fails**, another interface continues — **fault tolerance** for **real-time playback** such as Internet telephony. Example: client on two LANs, server on two networks → **four IP pairs**, but **current implementations** use **only one pair for normal communication**; the alternative is **failover only**. **SCTP does not currently allow load sharing** across paths. An association **allows multiple IPs per end**.

Full-duplex communication. Like TCP: data both ways at once; each end has **send and receive buffers**.

Connection-oriented service. Like TCP, but the connection is an **association**. Site A and site B: **establish association, exchange data both ways, terminate**.

Reliable service. Like TCP, **acknowledgements** check safe arrival (error control is pointed to, not fully worked in this lecture).

Features: TSN, SI, SSN, packets vs segments Transmission Sequence Number (TSN). TCP’s data unit is a **byte**, numbered with a **sequence number**. SCTP’s data unit is a **data chunk**, which **may or may not** be 1:1 with an application message because of **fragmentation**. Transfer is controlled by numbering **chunks**. **TSN** plays TCP’s sequence-number role: **32 bits, randomly initialized in 0 ... $2^{32}-1$** . Every data chunk **must carry its TSN** in its header.

Stream Identifier (SI). TCP: one stream per connection. SCTP: **several streams per association**. Each stream needs a **16-bit SI starting from 0**. Every data chunk **carries SI** so it can be placed in the right stream.

Stream Sequence Number (SSN). Delivery is **to the proper stream, in order**. Besides SI, each data chunk in a stream has an **SSN** to order chunks **inside that stream**.

Packets. TCP **segments** mix data bytes with **six header flags**. SCTP is different: **data chunks** and **control chunks**; **several of each can share one packet**. An SCTP **packet plays the role of a TCP segment**.

Differences the lecture lists:

TCP segment	SCTP packet
Control is in the header	Control is control chunks of several types
Data is one entity	Several data chunks , possibly different streams
Optional options field	No options field ; new features = new chunk types
Mandatory header 20 bytes	General header 12 bytes
Sequence number in header	TSN lives in each data-chunk header
ACK and window in header	ACK and window in control chunks
Header length (HL) needed because options vary	Fixed 12-byte header, no HL
Urgent pointer	None in SCTP
Checksum 16 bits	Checksum 32 bits
Connection ID = IP + ports	Multihoming = verification tag identifies the association
One sequence number = first data byte	Several chunks = TSN / SI / SSN per data chunk
Some control segments consume a sequence number	Control chunks never use TSN, SI, or SSN — those three belong only to data chunks , not the whole packet

Control and data travel in **separate chunks**. An association sends **many packets**; a packet may hold **several chunks**; chunks may belong to **different streams**.

Worked packing example. Process A sends **11 messages** to B on **three streams**: first **four** in stream 0, next **three** in stream 1, last **four** in stream 2. Assume each message fits **one data chunk** (no fragmentation) and the process delivers **stream 0, then 1, then 2**. Network allows **only three data chunks per packet** → **four packets**:

- Stream 0 chunks: packet 1 and part of packet 2
- Stream 1: rest of packet 2 and packet 3
- Stream 2: rest of packet 3 and packet 4

Each data chunk needs **three identifiers**: **TSN** (cumulative, for **flow and error control**), **SI** (which stream), **SSN** (order in that stream, starting at **0** per stream).

Acknowledgements. TCP ACKs are **byte-oriented** (sequence numbers). SCTP ACKs are **chunk-oriented** (they refer to **TSN**). TCP control-only segments still use seq/ACK (e.g. **SYN** needs **ACK**). SCTP **control chunks have no TSN**; they are **acked by another control chunk of the right type** — **INIT** by **INIT ACK**. **No sequence/ACK number is needed** for that. **SCTP acknowledgement numbers ack only data chunks**; control chunks ack **other control chunks** if needed.

Packet format Mandatory **12-byte general header** plus a **variable-length set of chunks**. **Control chunks** maintain the association; **data chunks** carry user data. In a packet, **control chunks come before data chunks**.

General header — four fields; defines endpoints, binds the packet to an association, and protects integrity **including the header**:

Field	Size	Role
Source port	16 bits	Sending process
Destination port	16 bits	Receiving process
Verification tag	(association ID)	Matches packet to this association; stops a packet from a previous association being accepted; repeated in every packet; separate tag each direction
Checksum	32 bits	CRC-32 (increased from TCP's 16 bits to allow CRC-32)

Chunks share a layout: first **three fields common**, then type-specific information. Information must be a **multiple of 4 bytes**; else **padding**. Chunks end on a **32-bit boundary**.

Common field	Size	Role
Type	8 bits	Up to 256 chunk types; few defined, rest reserved
Flags	8 bits	Meaning depends on type
Length	16 bits	Total chunk size in bytes , including type, flags, and length

Association: four-way handshake SCTP is connection-oriented; the connection is an **association** to emphasize **multihoming**. Establishment is a **four-way handshake**. Normally a **client** initiates toward a **server** that is **prepared to receive** associations (like TCP listen).

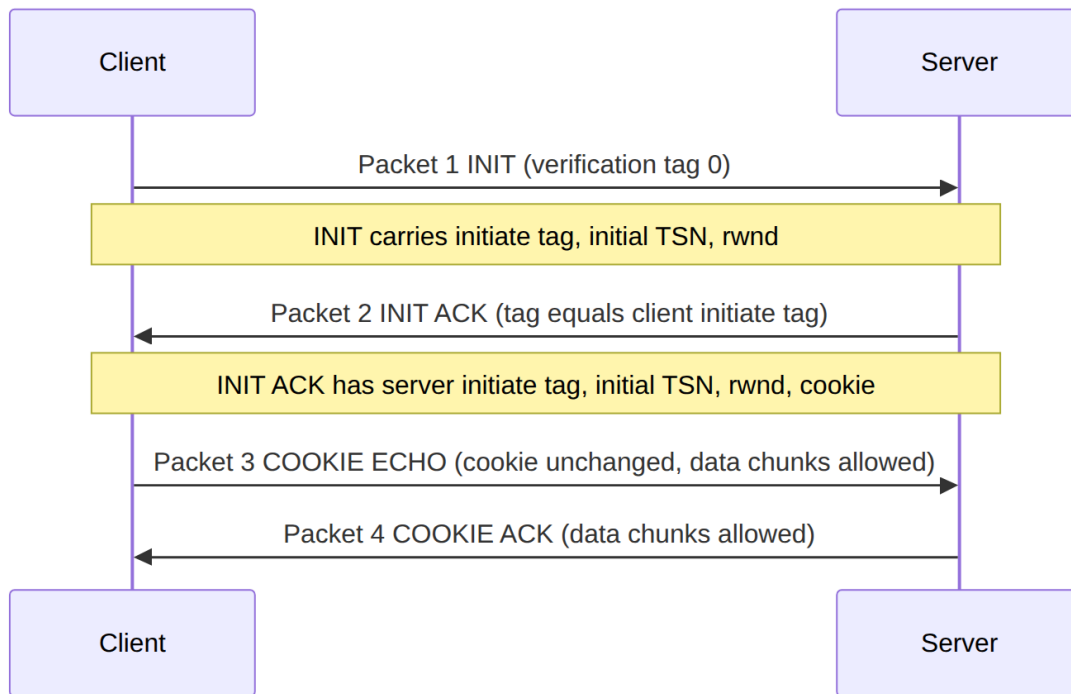


Figure 5: Architecture diagram

Normal steps

1. **Client → server: INIT chunk.** General-header **verification tag is 0** (none defined yet client→server). INIT includes an **initiate tag** for packets **server→client**, the **initial TSN** this direction, and advertises **rwnd** (receive window). **rwnd** is normally in a **SACK** chunk; it is sent here because SCTP allows **data chunks in packets 3 and 4**, so the server must know **client buffer size**. **No other chunks** with the first packet.
2. **Server → client: INIT ACK.** Verification tag = **initiate tag from INIT**. Chunk sets the tag for the **other direction**, **initial TSN** server→client, and **server rwnd** so the client may send a **data chunk in packet 3**. INIT ACK also carries a **cookie** defining the **server's state at that moment**.
3. **Client → server: COOKIE ECHO.** Simple chunk that **echoes the cookie unchanged**. **Data chunks are allowed** in this packet.
4. **Server → client: COOKIE ACK**, acknowledging COOKIE ECHO. **Data chunks allowed** here too.

(The lecture says cookie use would be discussed “shortly” and then ends after describing the echo; the taught fact is that the cookie **captures server state** and is **returned unmodified**.)

Key terms

Term	Meaning in this lecture
SCTP	Reliable, message-oriented transport combining UDP's boundaries with TCP's reliability, plus streams and multihoming
Association	SCTP's name for a connection (multihoming-aware)
TSN	32-bit transmission sequence number on data chunks
SI / SSN	Stream identifier and per-stream sequence number on data chunks
Chunk	Control or data block; several per packet; control before data
Verification tag	Per-direction association identifier (needed because of multihoming)
CRC-32	32-bit packet checksum in the 12-byte general header
INIT / INIT ACK / COOKIE	Four-way handshake chunks
ECHO / COOKIE ACK	
rwnd	Advertised receive window (in INIT/INIT ACK so data can ride packets 3-4)

Lecture takeaways

- SCTP exists because **SS7-style signaling and IP telephony** need **message boundaries and reliability together**, which neither UDP nor TCP fully gives.
- An **association** can carry **many streams** (head-of-line blocking on one stream does not freeze the others) and **many IPs** (failover; **not** load-sharing in “current” implementations).
- Numbering is **chunk-based**: **TSN** globally, **SI+SSN** per stream; **control never consumes TSN**.
- The on-wire unit is a **packet**: 12-byte header (ports, verification tag, CRC-32) plus padded chunks.
- Setup is **four-way** with a **cookie**, and **user data may already appear** in the third and fourth packets.

M29: ATM Network Security Protocol

Source: <https://www.youtube.com/watch?v=JdRurgWuHvQ> **Instructor / expert:** Dr. Lal Chand, Department of Computer Engineering, Punjabi University, Patiala. Course coordinator: Dr. Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala.

Learning objectives

- Define ATM as **asynchronous time-division multiplexing** of **fixed 53-byte cells** (5-byte header + 48-byte payload) on a **SONET/ISDN-style** backbone,

with a VC set up **before** data.

- Decode **UNI vs NNI** headers (GFC, VPI, VCI, PT/EFCI/AUU, CLP, HEC) and name ABR/CBR/UBR/VBR.
- List ATM Forum-style security goals (confidentiality, AAL integrity, signaling signatures, availability functions) and ATM-specific threats (**VC stealing**, traffic analysis).
- Recite QoS/traffic parameters (PCR, SCR, CLR, CTD, CDV, BT/MBS, MCR), CAC, and the lecture's ATM **disadvantages**.

Core concepts

Asynchronous Transfer Mode (ATM) is a **switching technique** for telecommunication networks that uses **asynchronous time-division multiplexing** to encode data into **small fixed-size cells**. That differs from **Ethernet or the Internet**, which use **variable-size packets/frames**. ATM is called the **core protocol over the SONET backbone of ISDN**.

Why cells, not variable packets ATM was designed with **cells** because **voice** is packetized and must **share a network with bursty data**. However small the voice packets, they would otherwise meet **full-size data packets** and suffer **maximum queueing delay**. So **all data units should be the same size**.

Fixed cells can be **switched in hardware** without delays of **routed frames and software switching** — why some people thought ATM was the key to the **Internet bandwidth problem**.

ATM **creates fixed routes between two points before transfer**, unlike **TCP/IP**, where packets may take **different routes**. That makes **usage accounting** easier, but an ATM network is **less adaptable to a sudden traffic surge**.

ATM provides **data-link services on OSI layer-1 physical links**. It behaves like a mix of **small packet switching and circuit switching**, suited to **real-time low-latency traffic (VoIP and video)** and **high-throughput transfers (files)**. A **virtual circuit (connection)** must exist before endpoints exchange data.

Bit-rate service classes

Class	Lecture meaning
Available bit rate (ABR)	Guaranteed minimum capacity; may burst higher when the network is idle
Constant bit rate (CBR)	Fixed rate, steady stream — analogous to a leased line
Unspecified bit rate (UBR)	No throughput guarantee ; for delay-tolerant apps such as file transfer
Variable bit rate (VBR)	Specified throughput but not sent evenly — popular for voice and video conferencing

(The transcript's first class is unnamed "bit rate provides a guaranteed minimum"; that is the **ABR** definition in this list.)

ATM cell structure An ATM cell is a **5-byte header** plus a **48-byte payload**. Two formats: **UNI (User-Network Interface)** and **NNI (Network-Network Interface)**. **Most links use UNI.**

Field	Size	Notes
GFC (Generic Flow Control)	4 bits (UNI)	Default 0 . Intended for local flow control / sub-multiplexing so several terminals can share one network connection, like two ISDN phones on one BRI . All four GFC bits must be zero by default
VPI (Virtual Path Identifier)	8 bits UNI / 12 bits NNI	Path identifier
VCI (Virtual Channel Identifier)	16 bits	Channel identifier
PT (Payload Type)	3 bits	Special cell kinds
CLP (Cell Loss Priority)	1 bit	Which cells to drop first in congestion
HEC (Header Error Control)	8 bits	CRC polynomial $x^8 + x^2 + 1$

PT field (user vs management)

- **PT bit 3 (MSB): 1 = network management cell; 0 = user data cell**, then:
 - **PT bit 2: Explicit Forward Congestion Indication (EFCI)** — **1** = congestion experienced
 - **PT bit 1: AUU** (ATM user-to-user) bit, used by **AAL5** to mark **packet boundaries**
- If MSB of PT is **1** (management), the other two bits distinguish: **network-management segment**, **network-management end-to-end**, **resource management**, and **reserved**.

ATM uses PT for **OAM** (Operations, Administration, and Management) cells and to **delineate packets** in some **ATM Adaptation Layers (AAL)**.

Several link protocols use **HEC** to drive a **CRC-based framing** algorithm that **finds cell boundaries with no extra overhead** beyond header protection. The 8-bit CRC **corrects single-bit header errors** and **detects multi-bit header errors**. On multi-bit errors, the **current and subsequent cells are dropped** until a cell with a **good header** is found.

NNI matches UNI except **GFC is reallocated to VPI**, extending VPI to **12 bits**. The lecture then claims a single NNI interconnection can address almost **2^{16} VPs**

of almost 2^{16} VCs each (field sizes taught are **12-bit VPI** and **16-bit VCI**); **some VP/VC numbers are reserved**.

ATM network interfaces

- **User-to-network interface** — public and private
- **Network-to-network / network-to-node interface** — **private NNI**, **public NNI**
- **Data Exchange Interface (DXI)** between **packet routers** and **ATM DSUs** (Digital Service Units)

ATM Adaptation Layer (AAL): how to **break application messages into cells**.

ATM layer: transmission, switching, reception, **congestion control**, **buffer management**, **cell-header generation/removal** at source/destination, **reset connection identifiers for the next hop** at a switch, **cell address translation**, **sequential delivery**.

ATM Forum security services **Data confidentiality** — protect business data from unauthorized persons; data available **only to intended, authorized** parties.

Data integrity and authentication — accuracy and consistency; data **not changed by unauthorized persons**. **AAL frames** are protected by appending a **cryptographic checksum**. **Reordering protection** is possible by putting a **sequence number into AAL frames before** that checksum.

Signaling protection — signaling may be **authenticated and integrity-protected** by a **digital signature** in an **information element (IE)**, especially if an **SME (security message exchange)** protocol is used. **SETUP and CONNECT** are protected by signing **IE fields specified by SME**. Protection is still offered for other messages (**RELEASE, STATUS, RESTART**) and for **SETUP/CONNECT if SME is not used**, by signing **part of the IE**.

Availability — the lecture restates integrity language here, then lists what an **ATM security system should provide**:

Function	Meaning
Verification of identities	Establish and verify claimed identity of any actor
Controlled access and authorization	No access to information/resources if not authorized
Protection of confidentiality	Stored and communicated data stay confidential
Protection of data integrity	Integrity of stored and communicated data
Strong accountability	An entity cannot deny its actions and their effects

Function	Meaning
Activities logging	Retrieve security-activity info in network elements, traceable to individuals/entities
Alarm reporting	Alarms for adjustable, selective security-related events
Audit	After violations, analyze security-relevant logs
Security recovery	Recover from successful or attempted breaches
Security management	Manage the services derived from the above; foundations of the security system . If the system cannot recover and stops providing security services, it is no longer secure . Security services and security information must themselves be managed securely

Threats to ATM Typical threats: **eavesdropping, spoofing, service denial, VC stealing, traffic analysis**. **VC stealing and traffic analysis happen only in ATM networks** (as taught).

Eavesdropping — tap the media and read data. Common. Because many ATM nets use **fiber**, people may **wrongly think** tapping is hard.

Spoofing — impersonate a user to a third party to use or destroy the victim's resources. May need **PDU-manipulation tools** and sometimes **special access** (e.g. **superuser on Unix**). Because networks interconnect via the **Internet**, it is **impossible to prevent** a hacker getting that access or even **tracing** who has it. ATM in the **public domain** is subject to this.

Service denial — ATM is **connection-oriented**; a VC is managed by **signals**. Setup uses **SETUP**; teardown uses **RELEASE** or **DROP PARTY**. If an attacker sends **RELEASE or DROP PARTY** to **any intermediate switch** on the VC, the VC **disconnects**. Doing that **frequently** disturbs communication and **disables QoS**. Combined with eavesdropping, the attacker can **block one user from another entirely**.

Stealing of VCs — if **two switches compromise**, they can steal a VC. Example: **VC1** (user U1) and **VC2** (user U2) both traverse switches **A** and **B**. **A** switches U1's cells A→B onto **VC2**; **B** switches them back to **VC1**. Switches forward on **VCI/VPI**; A and B **alter those fields back and forth**. Intermediate switches **will not notice** and treat cells as authentic VC2. In a public packet network U1 might gain little; in ATM with **guaranteed QoS**, U1 can steal a **higher-quality channel** they are **not entitled to**, and can gain more if **users pay for communication**. **U2 is hurt**. Switch compromise looks unlikely if **one organization** owns the net; in **ATM internetworking**, cells traverse **different ATM networks**, so two switches **compromising is much easier**.

Traffic analysis — infer information from **volume, timing, and parties** of a VC.

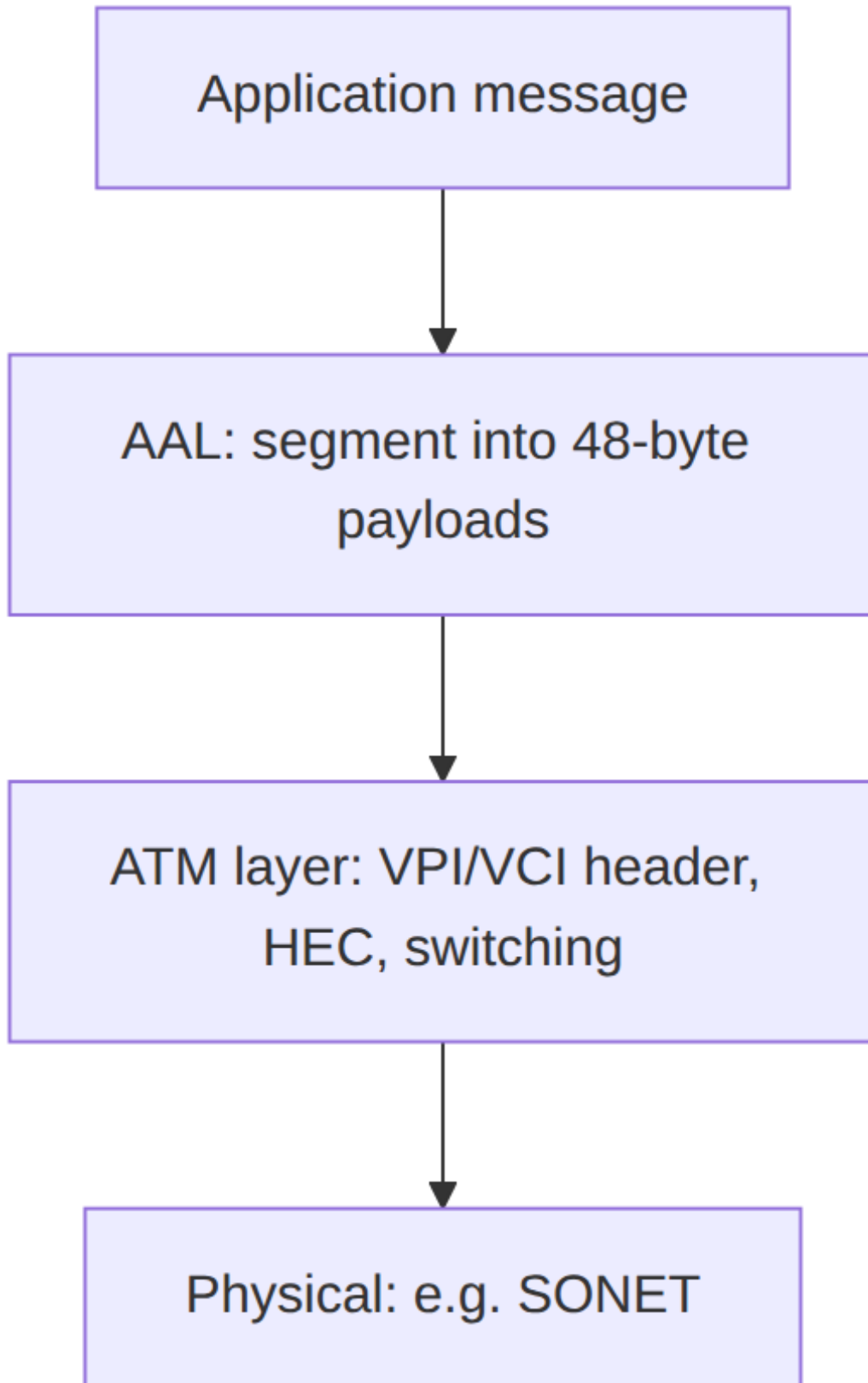


Figure 6: Architecture diagram

Volume and timing remain even if **data is encrypted**; **source and destination** often come from the **cell header in the clear** plus **routing-table knowledge**. Related: **covert channels** — encode information in **timing, volume, VCI, or even session keys** to leak data without being monitored. These two “won’t normally happen,” but may in **stringent-security** environments.

Traffic management and QoS parameters To deliver **guaranteed QoS on demand** while **maximizing utilization**, ATM needs **traffic management** almost everywhere: **signaling, routing, resource allocation, policing**.

Users can specify, at connection setup:

Parameter	Meaning
PCR (Peak Cell Rate)	Maximum instantaneous rate; inverse of minimum inter-cell interval (bursty traffic varies)
SCR (Sustained Cell Rate)	Average rate over a long interval
CLR (Cell Loss Ratio)	Percent of cells lost to error or congestion and not delivered. CLP=1 cells are discarded first in congestion; loss of CLP=0 is more harmful. CLR can be specified separately for CLP=1 and CLP=0
CTD (Cell Transfer Delay)	Delay from network entry to exit : propagation, queueing at switches, service times
CDV (Cell Delay Variation)	Variance of CTD; high variation ⇒ larger buffers for delay-sensitive voice/video
BT (Burst Tolerance)	Maximum burst that can be sent at peak rate ; bucket size of the leaky-bucket algorithm that controls traffic entering the network . Arriving cells go in a bucket drained at SCR ; MBS (Maximum Burst Size) is the max back-to-back cells at PCR
MCR (Minimum Cell Rate)	Minimum rate desired by a user

The **first six** were originally in **UNI version 3**.

Traffic contract. For guaranteed QoS, a contract at setup contains a **connection traffic descriptor** and a **conformance definition**. **Not every VC needs specified QoS**: if only specified-QoS connections were supported, a **large fraction of resources would be wasted** when connections **do not use their full contract**. **Unspecified QoS** can be supported **best-effort**, which is **enough for most existing data applications**.

Congestion control. Congestion is when **ingress to a link exceeds egress capacity**. Goal: **good throughput and delay with fair allocation**. One avoidance method: accept a new connection at setup **only if resources suffice** for acceptable QoS — **Connection Admission Control (CAC)**, required where QoS **must be guaranteed**.

Disadvantages of ATM (as taught) ATM was **not widely accepted**, though **some phone companies still use it in backbone networks**. **Expense, complexity, and lack of interoperability** blocked prevalence.

Drawback	Lecture detail
Expense	Even a moderate ATM switch costs far more than inexpensive LAN hardware ; ATM NICs cost much more than Ethernet NICs
Connection-setup latency	Connection-oriented paradigm: setup and teardown of a distant VC can take longer than using it
Cell tax	Headers impose a 10% tax on all data; Ethernet's comparable tax is 1%
No efficient broadcast	Connection-oriented nets are sometimes NBMA (Non-Broadcast Multiple Access) ; hardware lacks broadcast/multicast . Broadcast is simulated by an application copying data to each computer — inefficient
Complexity of QoS	Spec is cumbersome ; many implementations do not support the full standard
Assumption of homogeneity	Designed as a single universal system; minimal provision to interoperate with other technologies

Key terms

Term	Meaning in this lecture
ATM cell	5-byte header + 48-byte payload, hardware-switched
UNI / NNI	User-network vs network-network cell formats (GFC vs extra VPI bits)
VPI / VCI	Virtual path and channel IDs used for forwarding (and for VC-stealing attacks)
HEC	8-bit header CRC; framing plus single-bit correct / multi-bit drop
AAL / AAL5	Adaptation layer; AAL5 uses the AUU PT bit for packet boundaries
ABR / CBR / UBR / VBR	ATM bit-rate service classes
VC stealing	Compromised switches swap VPI/VCI to ride another user's QoS
CAC	Connection admission control at setup for guaranteed QoS
Cell tax	10% header overhead vs ~1% on Ethernet
NBMA	Non-broadcast multiple access — ATM has no native broadcast

Lecture takeaways

- ATM's security and performance story starts from **fixed cells and pre-established VCs** on a **SONET-class** fabric — good for **voice/video latency**, awkward for **surges** and **broadcast**.
 - Header fields (**VPI/VCI, PT, CLP, HEC**) are both the switching mechanism and the **attack surface** (clear-text parties, CLP drops, false RELEASE).
 - Forum-style security is **confidentiality, AAL checksums/sequence numbers, signed signaling, and a full management lifecycle** (log, alarm, audit, recover).
 - Unique ATM threats taught here: **VC stealing** between colluding switches and **traffic/covert-channel analysis** despite encryption.
 - QoS is a **traffic contract** (PCR/SCR/CLR/CTD/CDV/BT/MCR) plus **CAC**; unspecified QoS exists so the network is not wasted.
 - The closer is practical: ATM **lost the LAN** to cheaper, simpler, more interoperable Ethernet despite remaining in some **telco backbones**.
-

M30: Wireless Networks

Source: <https://www.youtube.com/watch?v=MdLxztyDzck> **Instructor / expert:** Dr. Navdeep Singh, Department of Computer Engineering, Punjabi University, Patiala. Course coordinator: Dr. Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala.

Learning objectives

- Define a wireless network as radio (OSI physical layer) used to avoid cabling, and classify **WPAN, WLAN, WMAN, WWAN**.
- Describe Bluetooth **IEEE 802.15**: piconet/scatternet, 2.4 GHz ISM FHSS/GFSK, baseband/LMP/L2CAP, and the 72/54-bit frame.
- Explain IEEE 802.11 BSS/ESS, station mobility, **DCF (CSMA/CA)** vs **PCF**, why CSMA/CD fails, 802.11 MAC frame/types, and **hidden vs exposed station** (RTS/CTS).
- Outline WiMAX (IEEE 802.16), WWAN microwave/cellular (cells, BSC/MSC), and satellite networks as taught.

Core concepts

The stated aims are to **identify basic topologies and their variations** and to **choose an appropriate topology** for a network plan. A **wireless network** is any computer network that uses **wireless data connections** between nodes. Homes, telecom networks, and enterprises use it to **avoid the cost of cabling** a building or linking equipment rooms. Wireless telecom is generally **radio**, implemented at the **OSI physical layer**.

Four categories: **Wireless PAN, Wireless LAN, Wireless MAN, Wireless WAN**.

Wireless PAN (Personal Area Network) WPANs interconnect devices in a **relatively small area**, generally **within a person's reach**. Examples: **Bluetooth radio** and **infrared light** linking a **headset to a laptop**.

Bluetooth Bluetooth transfers data between electronic devices over a **short distance** compared with other wireless modes. It **removes cords, cables, and adapters**. A Bluetooth LAN is an **ad hoc** network formed **spontaneously**: gadgets find each other and form a **piconet**. It can connect to the **Internet** if one gadget has that capability.

The name comes from **Harald Bluetooth (Harald Blåtand)**, king of Denmark who united **Denmark and Norway**; *Blåtand* translates as Bluetooth. Today Bluetooth implements **IEEE 802.15**. Protocols let devices **find and connect (pairing)** and **securely transfer data**.

Two network types: piconet and scatternet.

A **piconet** is a small net of **up to eight stations: one primary (master)** and the rest **secondaries (slaves)**. All secondaries **synchronize clocks and hopping sequence** with the primary. **Only one primary**. Communication is **one-to-one or one-to-many**. A piconet can have a **maximum of seven active secondaries; additional secondaries** (the lecture says **eight**) can be in the **parked state: synchronized with the primary but not communicating** until moved out of park. Because only **eight stations can be active**, activating a parked station means an **active station must park**.

A **scatternet** combines piconets: a **secondary in one piconet can be primary in another**, receiving from the first primary and forwarding to secondaries in the second piconet. A station can be a **member of two piconets**.

Radio: 2.4 GHz ISM band, 79 channels of 1 MHz. Physical layer: **frequency-hopping spread spectrum (FHSS)** to avoid interference. Bluetooth **hops 1600 times per second** (modulation frequency changes 1600/s). Bits map with **GFSK** (FSK with **Gaussian bandwidth filtering**): bit **1** = deviation **above** the carrier, bit **0** = deviation **below**. Carrier frequencies: **$F_c = 2402 + n$ MHz** for **$n = 0 \dots 78$** . Channel 0 uses **2402 MHz**, channel 1 **2403 MHz** (transcript "242/243 MHz" drops a zero).

Stack (as taught)

- **Link control / baseband** — analogous to a MAC sublayer but with physical-layer elements: how the **master controls time slots** and how slots **group into frames**.
- **Link Manager (LMP)** — logical channels, **power management, pairing, encryption, QoS**; sits **below the Host Controller Interface (HCI)**. Typically protocols **below HCI run on the Bluetooth chip, above HCI on the host device**.
- **L2CAP** (Logical Link Control and Adaptation Protocol) — **variable-length messages** and **reliability if needed**. Users include **SDP (Service Discovery Protocol)** to locate services, and **RFCOMM** which **emulates a PC serial port** (keyboard, mouse, modem).

- **Applications / profiles** — vertical slices of the stack for a purpose (e.g. **headset profile**). Profiles may include L2CAP if they send packets, or **skip L2CAP** for a **steady flow of audio samples**.

Bluetooth frame

Field	Size	Content
Access code	72 bits	Sync bits and primary identifier , to distinguish one piconet's frames from another
Header	54 bits	A repeated pattern; six subfields (see below). Implemented as three identical 18-bit sections
Payload	0-2740 bits	Upper-layer data or control

Header subfields:

Subfield	Bits	Role
Address	3	Up to 7 secondaries ; 0 = broadcast from primary to all secondaries
Type	4	Type of upper-layer data
F (flow)	1	Set $\Rightarrow$ device cannot receive more (buffer full)
A (ACK)	1	Stop-and-wait ARQ ; one bit suffices
S (seq)	1	Stop-and-wait sequence ; one bit suffices
HEC	8	Checksum over each 18-bit header section

The header's **three identical 18-bit copies** are compared **bit by bit**; if bits differ, **majority vote**. That is **forward error correction**. **Double error control** is needed because radio is **very noisy**. **No retransmission in this sublayer**.

Applications listed: wireless **phone headset** (phone in a bag; reduces **radiation to the head**); **PDA/PC/laptop sync**; send email from a laptop **via the phone after a flight**; **wireless mouse and keyboard**; phone **alert when the laptop gets mail**; **find a printer** from a laptop.

Key Bluetooth features: less complication, **less power**, **cheaper**, **robustness**.

Wireless LAN WLANs are the **most important access-network technologies** on the Internet. The most popular is **IEEE 802.11**, also known as **Wi-Fi**.

IEEE 802.11 defines two service sets:

Basic Service Set (BSS) — building block: stationary or mobile wireless stations plus an **optional central base station**, the **access point (AP)**.

- **BSS without AP: standalone**, cannot send data to another BSS — **ad hoc**. Stations **locate one another** and agree to form a BSS.
- **BSS with AP: infrastructure** network.

When **multiple BSSs** connect: stations in range can talk **without an AP**, but communication **between stations in different BSSs usually goes via two APs** — like **cellular**, if each BSS is a **cell** and each AP a **base station**. A mobile station can **belong to more than one BSS at once**.

Extended Service Set (ESS) — interconnection of BSSs (implied by the three mobility types).

Station types by mobility (IEEE 802.11)

Type	Movement
No-transition	Stationary, or moving only inside one BSS
BSS-transition	From one BSS to another, still inside one ESS
ESS-transition	From one ESS to another; IEEE 802.11 does not guarantee continuous communication during the move

MAC: DCF and PCF

IEEE 802.11 defines two MAC sublayers: **Distributed Coordination Function (DCF)** and **Point Coordination Function (PCF)**.

DCF uses **CSMA/CA** (carrier-sense multiple access with **collision avoidance**). WLANs **cannot implement CSMA/CD** for three reasons:

1. Collision detection would require **sending and receiving collision signals at once** → **costly stations** and **more bandwidth**.
2. Collisions may be **missed** because of the **hidden-station problem**.
3. **Distance and fading** can hide a collision at the far end.

PCF is **optional, infrastructure only** (not ad hoc), implemented **on top of DCF**, mostly for **time-sensitive** traffic. It is **centralized, contention-free polling**: the **AP polls** capable stations one after another; they send data **to the AP**. To **prefer PCF over DCF**, extra interframe spaces **PIFS** and **SIFS** are defined. **SIFS** is the same as in DCF; **PIFS is shorter than DIFS**, so if a DCF station and an AP both want the medium, the **AP wins**.

IEEE 802.11 MAC frame (nine field groups as taught)

Field	Size	Role
Frame Control (FC)	2 bytes	Frame type and control
Duration / ID	(in FC discussion)	In almost all types: duration of transmission used to set NAV ; in one control frame: frame ID
Address (×4)	6 bytes each	Meaning depends on To DS and From DS bits
Sequence control		Sequence number for flow control
Frame body	0-2312 bytes	Depends on FC type/subtype
FCS	4 bytes	CRC-32

Three frame categories: **management** (initial communication between stations and APs), **control** (channel access and acknowledgements), **data** (data and control information).

Hidden station problem Station **B**'s range is one oval, **C**'s another. **C is outside B's range and vice versa**. **A** hears **both**. If **B is sending to A**, **C** cannot hear **B**, thinks the medium **idle**, sends to **A** → **collision at A**. **B** and **C** are **hidden from each other with respect to A**. Hidden stations **reduce capacity**.

Solution: handshake frames (RTS/CTS). **RTS** from **B** reaches **A** **not C**. **CTS** from **A** (with **duration** of **B→A** data) reaches **both B and C** because both are in **A's range**. **C** then **refrains** until that duration ends.

Exposed station problem Inverse: **A transmits to B**. **C** has data for **D** that would **not interfere**, but **C hears A** and **refrains** — **too conservative**, wasting capacity. **RTS/CTS does not fully fix this**. **C** hears **RTS from A** but **not CTS from B**. **C** can wait for **B's CTS** to reach **A**, then send **RTS to D**. **A** (sending, not receiving) may ignore it; **B might CTS**. If **A has already started data**, **C cannot hear D's CTS** because of collision, so **C stays exposed until A finishes**.

Wireless MAN A **WMAN** connects several **WLANs**. **WiMAX** is a **WMAN** described by **IEEE 802.16**. It enables **wireless transmission of packets at broadband rates**, giving computers/mobiles **mobility and high-speed Internet** without cable or a **Wi-Fi hotspot**.

Implementation needs **telecom-scale infrastructure** like **GSM/CDMA: base stations, sectorized antennas, control centers**, and other critical parts.

Media claims of **over 30 miles** from **one base station** hold only in **ideal conditions**. Practically, **satisfactory broadband** is about **4-5 miles**; with **line of sight** up to **10 miles**. Coverage and QoS otherwise depend on **terrain and population**.

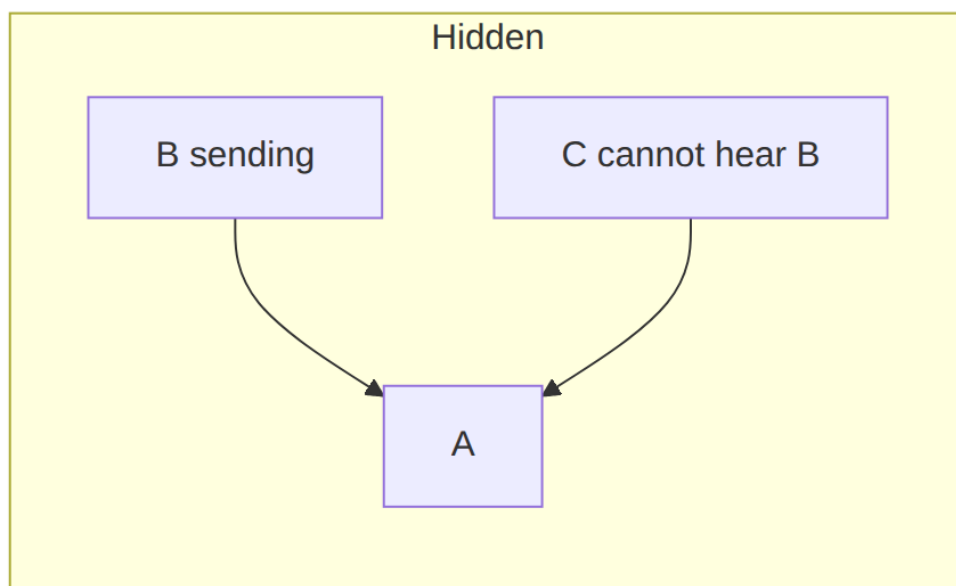
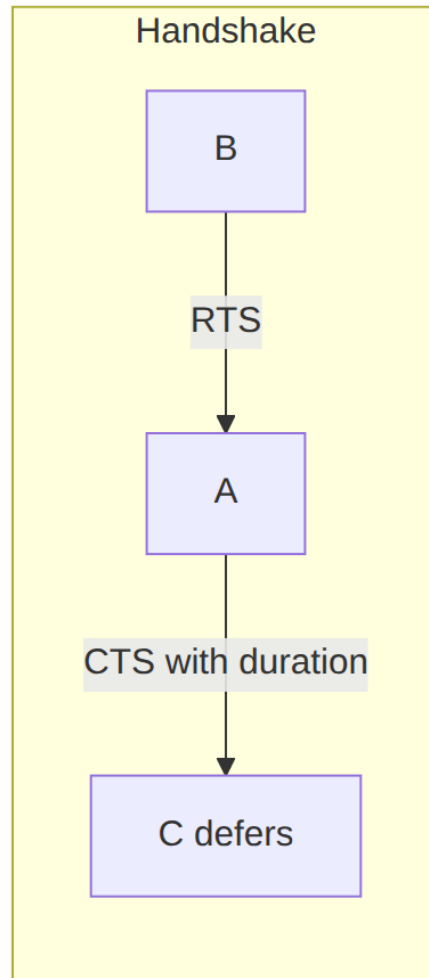


Figure 7: Architecture diagram

Wireless WAN WWANs cover **large areas** (neighboring towns/cities): **branch offices** or **public Internet access**. Links between access points are usually **point-to-point microwave** using **parabolic dishes** on the **2.4 GHz** band, not only the directional antennas of smaller nets. A typical system has **base-station gateways, access points, and wireless bridging relays**. Other configurations are **meshes** where each AP **relays**.

Wireless is also used in **cellular telephony and satellite networks**.

Cellular: communication between **two mobile stations** or between a mobile and a **stationary land unit**. The provider must **locate and track** a caller, **assign a channel**, and **hand the channel from base station to base station** as the caller leaves range. Each service area is split into **cells**. Each cell has an **antenna** and a **solar- or AC-powered base station**. Base stations are controlled by a **Mobile Switching Center (MSC)** that coordinates them with the **telephone central office: connects calls, records call information, bills**.

Cell size is not fixed: high-density areas need **more, geographically smaller cells**. Once set, size is optimized to **limit adjacent-cell interference**. **Transmit power is kept low** so a cell does not interfere with others.

Satellite network: nodes including **satellites** providing Earth-to-Earth communication. A node may be a **satellite, earth station, or end-user terminal/telephone**. A **natural satellite (the Moon)** could relay, but **artificial satellites** are preferred so we can install electronics that **regenerate** a weakened signal. Natural satellites are also **too far**, causing **long delay**. Like cellular, satellite nets **divide the planet into cells**. They can reach **any location**, however remote — **high-quality communication for undeveloped regions without huge ground infrastructure**.

The lecture closes: topologies and variations, wireless standards and features, and advantages; **security management is deferred to the next lecture**.

Key terms

Term	Meaning in this lecture
WPAN / WLAN / WMAN / WWAN	Personal, local, metro, and wide-area wireless scopes
Piconet / scatternet	Bluetooth 1-master small net vs overlapping piconets
Parked state	Bluetooth secondary synced but not in the active seven
GFSK / FHSS 1600 hop/s	Bluetooth 2.4 GHz ISM modulation and hopping
BSS / ESS / AP	802.11 building block, extended set, optional access point
DCF / PCF	CSMA/CA distributed access vs AP polling (PIFS < DIFS)
Hidden / exposed station	Collision because sender cannot hear a peer vs false deferral
RTS/CTS	Handshake that advertises duration so hidden nodes defer

Term	Meaning in this lecture
WiMAX / IEEE 802.16	Broadband WMAN
MSC	Mobile Switching Center in cellular

Lecture takeaways

- Wireless is classified by **reach**: PAN (Bluetooth 802.15), LAN (Wi-Fi 802.11), MAN (WiMAX 802.16), WAN (microwave, cellular, satellite).
- Bluetooth is a **master-driven hopped piconet** (79×1 MHz, 1600 hop/s, GFSK) with a noisy-radio frame that **votes on a triple header** instead of retransmitting at baseband.
- 802.11 infrastructure is **BSS+AP**; ad hoc is BSS without AP. MAC is **CSMA/CA**, not CD, because of cost, **hidden nodes**, and fading.
- **RTS/CTS** mitigates hidden stations; the **exposed-station** problem is left as a capacity waste the handshake does not fully solve.
- Cellular **low-power cells + MSC** and satellites **cell-ize the planet** so coverage does not require cabling every site.

M31: Wi-Fi Security Protocol

Source: https://www.youtube.com/watch?v=Ml3_13Hn9-I **Instructor / expert:** Prof. Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar. Course coordinator: Dr. Maninder Singh, Professor and Dean of Academic Affairs, Thapar University, Patiala.

Learning objectives

- Distinguish **IEEE 802.11** (PHY/MAC specs) from the **Wi-Fi Alliance** brand, interoperability testing, and required **WPA/WPA2** plus **EAP**.
- Describe **FHSS vs DSSS** (chips), the 802.11 split into **PLCP/PMD**, MAC management, and **DCF (CSMA/CA) vs PCF**.
- Tabulate 802.11 / a / b / g / n / ac (and d, e, f, h, i, j, ad, af) with bands, modulation, rates, and ranges the lecture gives.
- Decode the 802.11 **MAC frame** and **Frame Control** bits, including the **WEP** body-encryption flag.

Core concepts

Despite the playlist title **Wi-Fi Security Protocol**, this lecture is an introduction to **Wi-Fi technology**: architecture, **IEEE 802.11 standards**, and **frame format**. Security appears as **Alliance certification (WPA/WPA2, EAP)**, **802.11i**, and the **WEP bit** in Frame Control. Outline: Wi-Fi intro, **layered protocol architecture**, **standards**, **frame format**.

WLAN vs Wi-Fi vs IEEE vs Alliance A **wireless LAN (WLAN)** is a data communication system used as an **extension or alternative to a wired LAN**. **Wi-Fi** is WLAN technology based on the **IEEE 802.11** series, issued by the **Institute of Electrical and Electronics Engineers (IEEE)**.

IEEE does not test equipment for compliance. The nonprofit **Wi-Fi Alliance** formed in **1999** to **establish and enforce interoperability and backward compatibility** and to **promote WLAN technology**. It restricts the **Wi-Fi brand** to 802.11-based technologies. Member manufacturers whose products follow 802.11 may mark them with the **Wi-Fi logo**.

Certification requires compliance with:

- **IEEE 802.11 radio** standards
- **WPA and WPA2** security standards
- **EAP** authentication standards

Any manufacturer building to the standard may put a **Wi-Fi logo** on the product (as stated).

Spread spectrum on WLANs Most WLANs use **spread spectrum**: a **wideband RF** technique that **spreads the signal over the available bandwidth**. Two types:

Frequency-hopping spread spectrum (FHSS) — a **narrowband carrier** that **changes frequency in a pattern known only to transmitter and receiver**.

Direct-sequence spread spectrum (DSSS) — generates a **redundant bit pattern for each bit**, needing **more bandwidth**. That pattern is a **chip / chipping code**. The receiver recovers the original even if **one or more chip bits are damaged**; **statistical techniques in the radio** recover data **without retransmission**.

Layered protocol architecture of IEEE 802.11 IEEE 802.11 covers only the **physical layer** and **medium-access layer** of OSI.

PHY split

- **PLCP (Physical Layer Convergence Protocol)** — provides **Clear Channel Assessment (CCA)** (carrier sense) and a **common PHY SAP independent of transmission technology**.
- **PMD (Physical Medium Dependent)** — **modulation and encoding/decoding**.

MAC management supports **medium access control**, **association and reassociation** of a station to an AP, **roaming** between APs, **authentication**, **encryption**, **synchronization** with an AP, and **power management** to save battery.

PHY modulations named

Technique	Lecture facts
FHSS	79 channels of 1 MHz ; a PRNG produces the hop sequence; a fair way to allocate the unregulated ISM band

Technique	Lecture facts
DSSS	1 or 2 Mbps ; technique similar to CDMA
OFDM	54 Mbps and above in the wider 5 GHz ISM band
HR-DSSS (high-rate DSSS)	11 and 22 Mbps in 2.4 GHz

MAC modes

DCF (Distributed Coordination Function) — **no central control**. Protocol: **CSMA/CA**. Both **physical** and **virtual** channel sensing.

- **Physical sensing:** station wants to send → senses. If **idle**, it **transmits** (does **not** sense while sending) and sends the **full frame**, which may be **destroyed at the receiver by interference**. On collision, wait a **random time** using Ethernet **binary exponential backoff**, then retry.
- **Virtual sensing:** **MACA** (Multiple Access with Collision Avoidance).

PCF (Point Coordination Function) — a **base station controls all activity in its cell**.

IEEE 802.11 standards IEEE 802.11 is a set of **MAC and PHY** specifications for WLANs in the **2.4 GHz and 5 GHz** bands (transcript “2.5 4 and 5”). Created and maintained by the **IEEE LAN standards committee**. **Base version 1997**, then many amendments. They are the basis for products using the **Wi-Fi brand**. Each standard suits a particular environment.

Standard	Released	Rate	Indoor / outdoor range	Spread / modulation	Band
802.11 (legacy)	1997	1-2 Mbps	20 m / 100 m	FHSS and DSSS	2.4 GHz ISM
802.11a	1999	54 Mbps (transcript “54 and 8”)	35 m / 120 m	OFDM	5 GHz ISM
802.11b	1999	11 and 22 Mbps	35 m / 120 m	HR-DSSS	2.4 GHz ISM
802.11g	2003	54 Mbps	38 m / 140 m	OFDM and DSSS	2.4 GHz ISM

Standard	Released	Rate	Indoor / outdoor range	Spread / modulation	Band
802.11n	2009	54-600 Mbps	70 m / 250 m	OFDM ; adds MIMO antennas	2.4 GHz and 5 GHz (5 GHz optional)
802.11ac	2013	about 1300 Mbps	(not separately ranged)	built on n	5 GHz ISM

802.11a note: 2.4 GHz is **heavily used**, so the **relatively unused 5 GHz** band is a **theoretical advantage**. **802.11a signals are absorbed more by walls** (smaller wavelength) and **penetrate poorly**.

Other 802.11 amendments

Amendment	Focus in this lecture
802.11d	Spread of the technology to countries not addressed by the base IEEE rules
802.11e	QoS for wireless multimedia
802.11f	Roaming between APs and interoperability between vendor groups
802.11h	Frequency selection and power on 5 GHz in European countries
802.11i	Enhancing WLAN security and authentication , including RADIUS, Kerberos, and IEEE 802.1X
802.11j	Japanese equivalent of 802.11h
802.11ad	New PHY in the 60 GHz spectrum; products under the WiGig brand. Certification later moved to the Wi-Fi Alliance (from the WiGig Alliance). Peak rate 7 Gbps
802.11af	Also White-Fi or Super Wi-Fi ; approved February 2014 . WLAN in TV white space, VHF/UHF 54-790 MHz , using cognitive radio on unused TV channels

Radio carriers, bands, and 2.4 GHz channels WLANs use **electromagnetic waves** without wires. Radio waves are **radio carriers**: they **deliver energy to a remote receiver**. Data are **superimposed** on the carrier so they can be extracted — **carrier modulation**.

Band use recap: **802.11b and 802.11g → 2.4 GHz**; **802.11a → more heavily regulated 5 GHz**; **802.11n → both**.

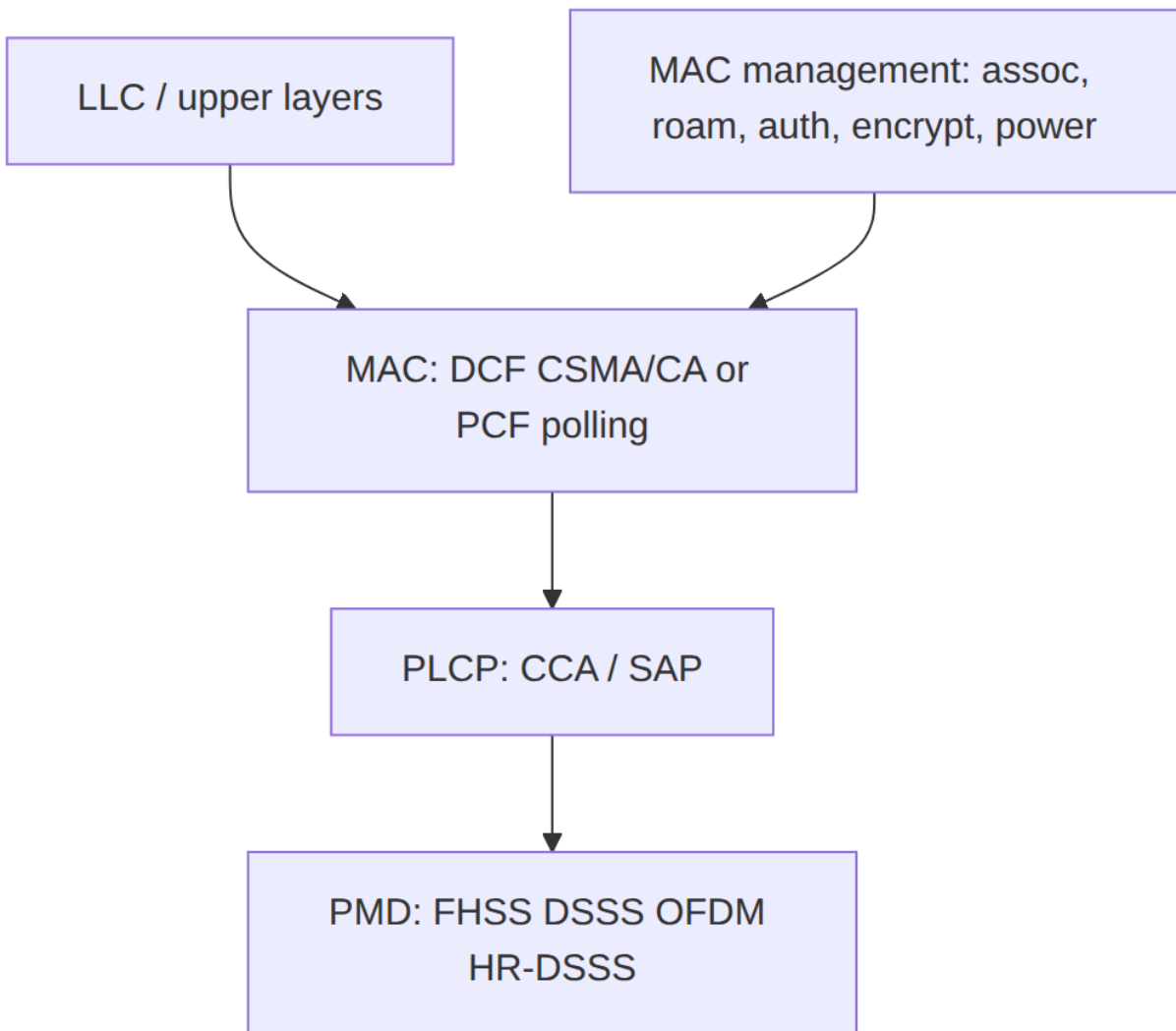


Figure 8: Architecture diagram

Each spectrum is subdivided into **channels** with a **center frequency and bandwidth**, like radio/TV. The **2.4 GHz band has 14 channels spaced 5 MHz apart**. **Channel 1** is centered on **2412 MHz** (transcript “2412 GHz”). Some channels have **extra restrictions or are unavailable** in some regulatory domains. The lecture shows a figure of all **14 channel frequencies**.

IEEE 802.11 frame format Nine fields from **Frame Control** through **checksum**. **Frame Control is 2 bytes with 11 subfields** (detailed after the main fields).

Field	Role
Frame Control	2 bytes, control (11 subfields)
Duration	How long this frame and its acknowledgement will occupy the channel
Four addresses	Source and destination plus source and destination base stations for intercell traffic
Sequence	Fragment numbering: of 16 bits , 12 identify the frame , 4 identify the fragment
Data (payload)	Up to 2312 bytes
Checksum / FCS	4 bytes , usual CRC

Frame Control subfields

Subfield	Meaning
Protocol version	Two protocol versions can run in the same cell
Type	Data, control, or management
Subtype	Further type detail
To DS / From DS	Frame going to or coming from the intercell distribution system (e.g. Ethernet)
MF	More fragments follow
Retry	Retransmission of a frame sent earlier
Pwr (power management)	Used by the base station to put the receiver to sleep or wake it
More	Sender has additional frames for the receiver
W (WEP)	Frame body encrypted with WEP (Wired Equivalent Privacy)
O (order)	Receiver must process a sequence of frames with this bit set strictly in order

Key terms

Term	Meaning in this lecture
Wi-Fi Alliance	1999 nonprofit that certifies 802.11 interoperability; requires WPA/WPA2 and EAP
FHSS / DSSS / OFDM / HR-DSSS	802.11 PHY families (hops, chips, 54 Mbps+ OFDM, 11/22 Mbps HR-DSSS)
PLCP / PMD	PHY convergence (CCA) vs medium-dependent modulation
CSMA/CA	DCF access: sense idle, send full frame, binary exponential backoff on collision; virtual sensing via MACA
MIMO	Multiple antennas added in 802.11n (54–600 Mbps)
IEEE 802.11i	Security/authentication amendment: RADIUS, Kerberos, 802.1X
WEP	Wired Equivalent Privacy; Frame Control W bit means the body is WEP-encrypted
802.11ad / WiGig	60 GHz PHY, peak 7 Gbps
802.11af / White-Fi	TV white-space WLAN, 54–790 MHz cognitive radio

Lecture takeaways

- **IEEE writes 802.11; the Wi-Fi Alliance owns the logo** and, for certification, insists on **WPA/WPA2 and EAP** — that is this lecture’s primary “security protocol” statement.
- 802.11 is **PHY + MAC only**: PLCP/PMD below, DCF/PCF above, with MAC management for **associate, roam, authenticate, encrypt, sleep**.
- Rate/range progression taught: **1-2 Mbps (1997) → a/b (1999) → g (2003) → n MIMO (2009) → ac ~1.3 Gbps (2013)**, plus **ad (60 GHz)** and **af (TV white space)**.
- **802.11i** is the amendment that names **RADIUS, Kerberos, and 802.1X**.
- The MAC header’s four addresses and Duration field support **distribution-system forwarding and channel reservation**; the **W bit** flags **WEP** on the body.

M32: Wi-Fi Security Protocol

Source: <https://www.youtube.com/watch?v=vCdk5ufJpPo> **Instructor / expert:** Dr. Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

Learning objectives

- State why IEEE 802.11 needed a confidentiality protocol and why WEP failed that role.

- Describe WEP's two authentication methods and the RC4/CRC/IV encryption pipeline.
- List the design flaws, IV weakness, and the mitigations that still left WEP crackable in minutes.
- Explain WPA as draft IEEE 802.11i (TKIP + Michael) and WPA2 as the full 2004 standard (AES).
- Compare WPA and WPA2 on 802.1X/EAP, PSK, mix-mode, cryptography, and processing cost.

Core concepts

Wireless networking is popular for home and business use, so many products and protocols exist. Radio is available to authorized users **and** to unauthorized users (hackers). IEEE **802.11** therefore offered a protection mechanism: **Wired Equivalent Privacy (WEP)** — a set of instructions and rules so that wireless data can travel over the air with some security.

Three protocols in this lecture The instructor covers three basic Wi-Fi security protocols, then compares the last two:

Protocol	Role in the lecture
WEP	Original 802.11 confidentiality algorithm (1997); soon replaced
WPA	Wi-Fi Protected Access — interim fix for known WEP issues
WPA2	Full IEEE 802.11i (2004); AES instead of TKIP

WEP was meant to provide **confidentiality** by encrypting traffic on the wireless network. It was **replaced by WPA** because of a flaw: WEP can be **cracked in a few minutes** with automated tools. WPA and WPA2 were designed to **address and fix** those known WEP issues.

WEP as originally specified (1997) WEP is a security algorithm for IEEE 802.11 WLANs, introduced with the original **802.11 standard ratified in 1997**. The intention was data confidentiality **comparable to a traditional wired network** — hence the name.

IEEE 802.11 design objectives for WEP:

Objective	Meaning as taught
Reasonably strong	Security should rest on the difficulty of discovering the secret key by brute force; that difficulty depends on key length and how often keys change

Objective	Meaning as taught
Self-synchronizing	WEP should resynchronize per message . That is critical for a data-link encryption algorithm that assumes best-effort delivery
Efficient	Implementable in hardware or software
Exportable	Designed to maximize chances of U.S. government export approval
Optional	Implementation and use in 802.11 should be optional

Those objectives show WEP was **not military-grade**. The intention was to make break-in **hard, not impossible**.

WEP authentication WEP security has two parts: **authentication** and **encryption**.

Authentication happens when a device **first joins the LAN**. The goal is to stop stations joining unless they know the **WEP key**. Two methods:

Open System Authentication

- The WLAN client **need not provide credentials** to the access point during authentication.
- Any client can authenticate with the AP and then attempt to associate.
- **In fact no authentication occurs.**
- Afterward, WEP keys **can** be used to encrypt data frames; the client must then have the **correct keys**.

Shared Key Authentication — a four-step challenge-response handshake:

1. The wireless device sends an **authentication request** to the AP.
2. The AP sends a **128-bit random authentication challenge in cleartext**.
3. The device uses the **shared secret key** to sign the challenge and returns an **authentication response**.
4. The AP decrypts the signed message with the same shared key and **verifies the challenge**. If it matches, authentication succeeds and the AP sends an **authentication success** message; otherwise it fails.

After authentication, no new secret key is exchanged. The **same shared key** is used for both authentication and encryption. There is therefore **no way to tell** whether a later message comes from the trusted device or from an **imposter**. This authentication is **prone to man-in-the-middle attack**.

WEP encryption (RC4 + CRC + IV) WEP uses the **RC4 stream cipher** between AP and device. As taught: WEP uses **8-bit RC4** and operates on 8-bit values by creating an array of **256 eight-bit values** as a lookup table (RC4's S-box).

Pipeline:

1. WEP uses **CRC** for integrity: a CRC is computed on the **plaintext** and **concatenated** to that plaintext.
2. The **secret key is concatenated with the initialization vector (IV)** and fed into RC4.
3. RC4 emits a **keystream** from that key and IV.
4. Keystream is **XORed** with (plaintext + CRC) to produce **ciphertext**.
5. The **same IV is prepended in cleartext** to the ciphertext.
6. **IV + ciphertext**, plus frame headers, go over the air.

Attack implications taught here:

- **Part of the secret is used with different exposed values.** An attacker can recover the secret by analyzing a portion of bits in the **first few bytes of the keystream**, with relatively little work.
- Concatenating the shared key with a **visible IV** is the **IV weakness**.

Why WEP was believed strong, then collapsed Early on, WEP was believed to offer **impenetrable resistance** to eavesdroppers and hackers. As WLANs grew, cryptanalysts found **flaws in the original design**. Many believe there was **little peer review**; those flaws would have been caught in design if specifications had been reviewed thoroughly.

For most users — especially **home users** — WEP was the **only choice** until new 802.11 mechanisms arrived. The instructor's line: **something is better than nothing**. Even with known weakness, WEP was more effective than **no security**, at least against unauthorized use and **eating up the bandwidth**.

Suggested improvements that were still not enough Proposed ways to paper over WEP:

- Choose a **bigger IV**.
- Prepend or append a **hash of the IV** to the ciphertext **instead of the IV in the clear**.
- Replace CRC with a stronger integrity check — **hash functions**.
- **Change the secret key regularly and dynamically** using secure symmetric-key distribution.
- **Better key management** using security handshake protocols.
- **New authentication** using **Extensible Authentication Protocol (EAP)**.

Even with those improvements, WEP still **cracked in a few minutes** with automated tools. Additional measures beyond WEP alone are required.

From WEP to WPA to WPA2 (IEEE 802.11i)

Year	Event
2003	Wi-Fi Alliance announced WEP superseded by WPA

The Alliance implemented 802.11i in **two steps**: **WPA**, then **WPA2**.

- **WPA** was an **intermediate** measure pending full 802.11i. It could be added by **firmware upgrades** on NICs designed for WEP — hardware that **could not support WPA2**. The interim software path **delayed buying new hardware**. Some old APs still needed **replacement or firmware upgrade**.
- WPA is sometimes called **draft IEEE 802.11i** (available **2003**).
- WPA2 became available in **2004**, commonly known as **full IEEE 802.11i** or **IEEE 802.11i-2004**.

WPA: TKIP and Michael WPA is designed to fix known WEP issues and give higher assurance that data stay protected, using **Temporal Key Integrity Protocol (TKIP)** for encryption.

	WEP	WPA (TKIP)
Key	40-bit or 104-bit key, manually entered on APs and devices, does not change	Per-packet key : dynamically generates a new 128-bit key for each packet
Integrity	CRC	Message integrity check — algorithm Michael

Per-packet keys are meant to stop the attacks that compromised WEP. The MIC is meant to stop an attacker **altering and resending** packets. CRC's main WEP flaw was that it did **not** provide a sufficiently strong integrity guarantee. Stronger **message authentication codes** existed but **required too much computation for old network cards**.

Michael is much stronger than CRC, but researchers still found a WPA flaw from Michael's limitations: retrieving the **keystream from short packets** for **re injection and spoofing**.

WPA2: AES and 802.11i features To solve those WPA problems and **fully implement IEEE 802.11i**, **WPA2** is used. Only authorized users should access the wireless device. Features taught: **stronger cryptography, stronger authentication, control, key management, replay-attack protection**, and **data integrity**.

Encryption is **AES**, not TKIP. As taught:

- Fixed **block size 128 bits**.
- Three key sizes, described as three iterations of the algorithm:
 - **128-bit** key → **9 rounds**
 - **192-bit** key → **11 rounds**
 - **256-bit** key → **13 rounds** (caption "133" is this figure)

- AES is a **substitution cipher**: within each round, bits are **substituted and rearranged**, then a **special multiplication** is performed on the new arrangement.
- Effectiveness: time to break AES-128 by brute force is given as around **2,200 years**.

WPA versus WPA2 Similarities

- Both use **802.1X**, i.e. the **EAP framework**, for **central mutual authentication** and **dynamic key management**.
- Both offer a **pre-shared key (PSK)** for **home and small-office** environments.
- Both are designed to secure all versions of 802.11 devices taught here: **802.11b**, **802.11a**, and **802.11g**.

Differences

- **WPA2 mix-mode** supports **both WPA- and WPA2-enabled devices on the same WLAN**; WPA supports **only one**.
- Most significant: **WPA2 uses AES instead of TKIP**.
- **WPA2 is theoretically not hackable; WPA is**.
- **WPA2 requires more processing power than WPA**.

Diagrams

Key terms

Term	Meaning in this lecture
WEP	Wired Equivalent Privacy — original 802.11 confidentiality algorithm (1997)
WPA	Wi-Fi Protected Access — 2003 interim / draft 802.11i using TKIP
WPA2	Full IEEE 802.11i-2004 using AES
IEEE 802.11i	WLAN security standard implemented in two steps as WPA then WPA2
Open System Authentication	Join path that does not actually authenticate; encryption keys applied later
Shared Key Authentication	Four-step WEP challenge-response using the same key later used to encrypt
RC4	Stream cipher WEP uses with a 256-entry 8-bit lookup table
IV	Initialization vector concatenated with the WEP key; sent in the clear — the IV weakness
CRC	WEP integrity check on plaintext; too weak to stop alteration
TKIP	Temporal Key Integrity Protocol — per-packet 128-bit keys in WPA
Michael	WPA message integrity check; stronger than CRC, still flawed on short packets
AES	WPA2 block cipher: 128-bit blocks; 128/192/256-bit keys; 9/11/13 rounds as taught
802.1X / EAP	Framework both WPA and WPA2 use for mutual authentication and dynamic keys

Term	Meaning in this lecture
PSK	Pre-shared key mode for home / small office
Mix-mode	WPA2 ability to host WPA and WPA2 stations on one network

Lecture takeaways

- Air is shared with hackers; 802.11's first answer was WEP, designed to be **hard** to break and **optional**, not military-grade.
- WEP's shared-key handshake reuses one key for auth and encryption and is **MITM-prone**; RC4 with a **clear IV** and **CRC** leaked keystream.
- Firmware-friendly **WPA/TKIP/Michael** was only a bridge to **802.11i**; Michael still allowed short-packet reinjection.
- **WPA2/AES** is the full 2004 standard: stronger crypto, authentication, key management, replay protection, and integrity — at higher CPU cost.
- Prefer WPA2; WPA is described as hackable, WPA2 as theoretically not. Both still share 802.1X/EAP, PSK, and 802.11a/b/g coverage.

M33: Bluetooth Networks and Security Protocols

Source: https://www.youtube.com/watch?v=wzD1F-Knr_E **Instructor / expert:** Prof. Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

Learning objectives

- Place Bluetooth as a short-range UHF ISM WPAN technology and recount SIG / IEEE 802.15 origins.
- Draw a piconet and scatternet and state the master/slave communication rule.
- Walk the Bluetooth protocol stack, ACL vs SCO links, L2CAP roles, and the 54-bit header frame.
- Recite SIG versions from 1.0 through 4.2 and the IEEE 802.15.x WPAN family.
- Name Bluetooth's three native security services and the four BR/EDR/HS security modes, including NIST's warning on mode 1.

Core concepts

Bluetooth is a wireless standard for exchanging data over **short distances** using **short-wavelength UHF** radio. It operates in the **ISM band from 2.4 to 2.485 GHz**, for mobile devices and **personal area networks (PANs / WPANs)**.

Origins and naming In **1994**, **Ericsson** wanted to connect mobile devices to other devices **without cables**. With **IBM, Intel, Nokia, and Toshiba** it formed a **Special Interest Group (SIG)**. They developed a short-range, **low-power, inexpensive** radio standard for computing and communication devices and accessories.

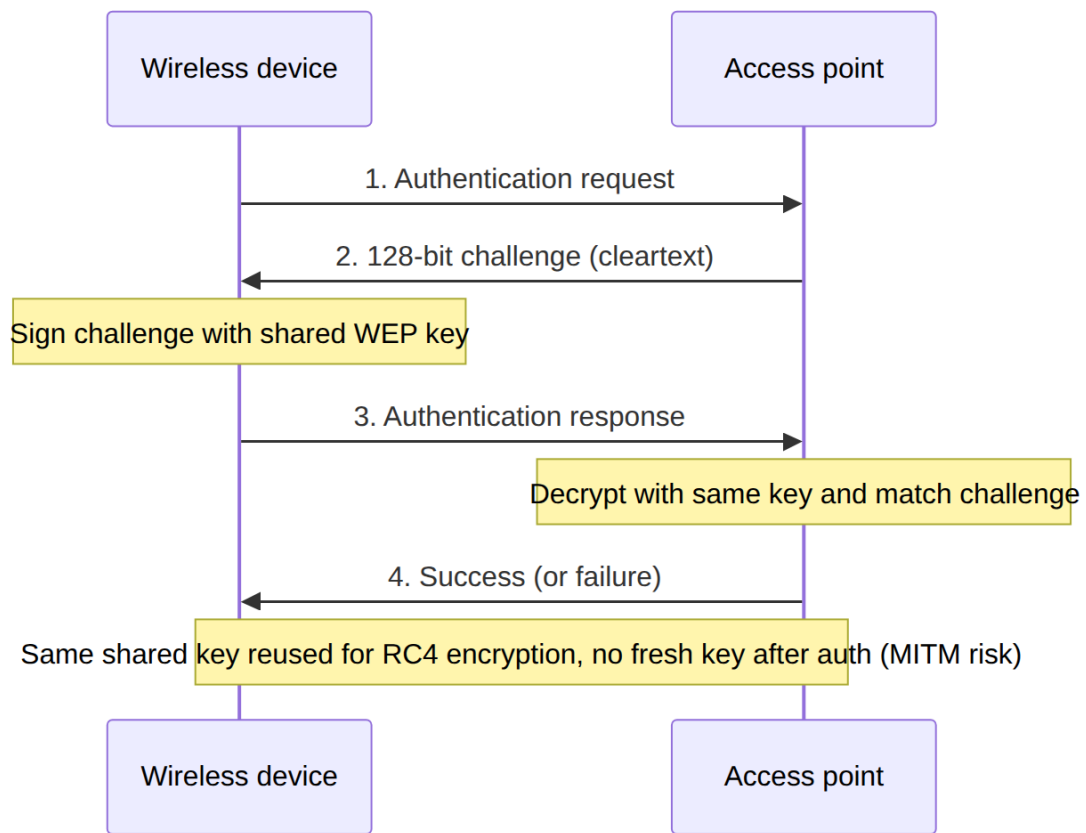


Figure 9: Architecture diagram

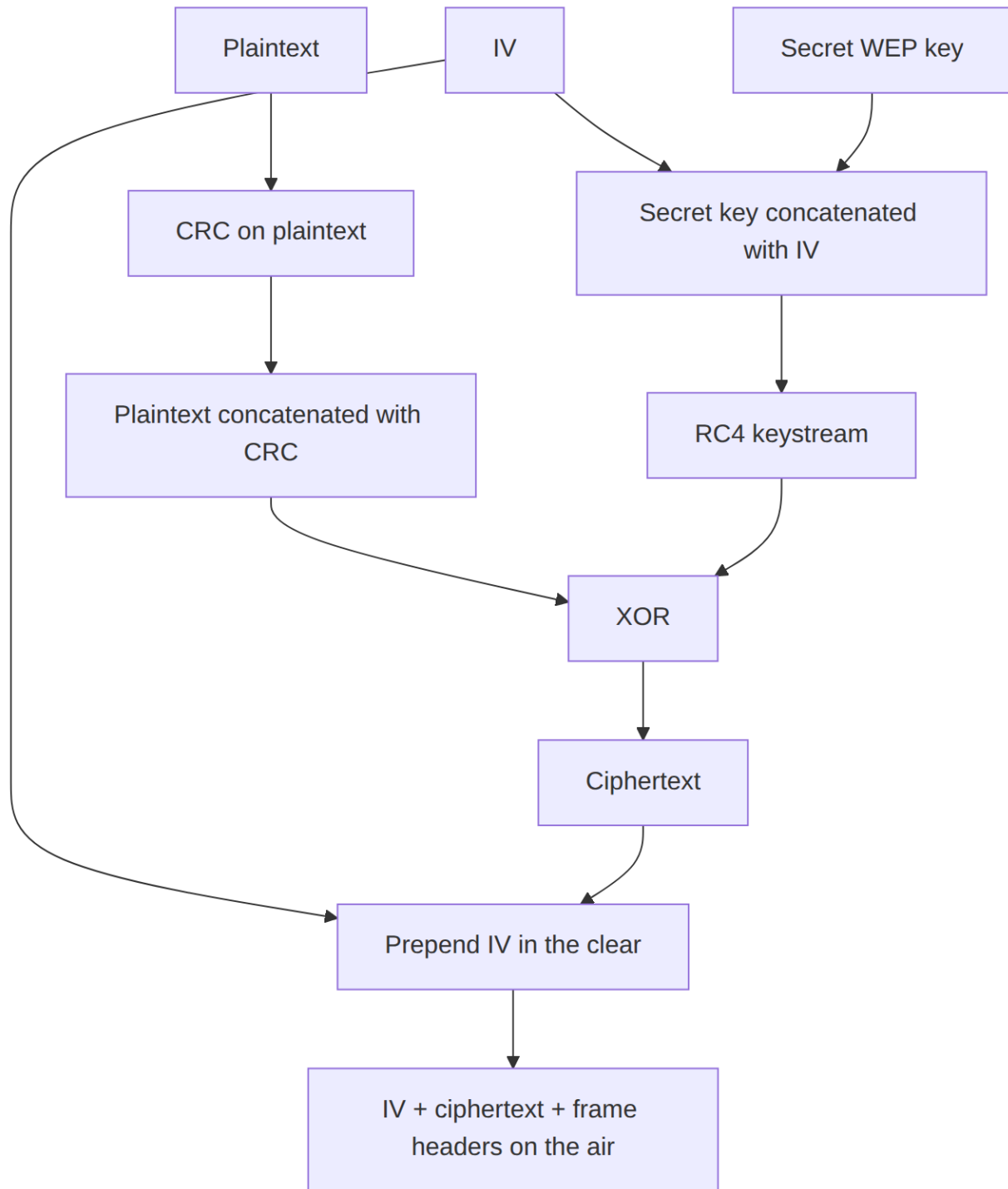


Figure 10: Architecture diagram

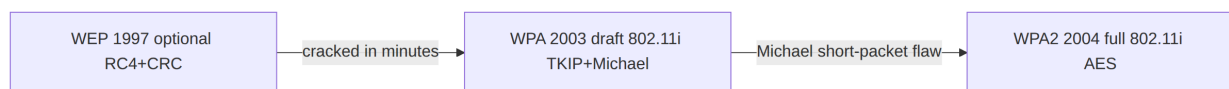


Figure 11: Architecture diagram

The name **Bluetooth** is after the 10th-century king **Harald Bluetooth**, who united distant Danish tribes into one kingdom. The name was proposed in **1997** by **Jim Kardach**, who had developed a system allowing mobile phones to talk to computers.

The Bluetooth SIG defines manufacturing standards. In **July 1999** it issued a **1,500-page specification of version 1.0**. Shortly after, the IEEE WPAN working group **802.15** adopted that document as a basis.

- **Bluetooth SIG specification:** a **complete system**, physical layer through application layer.
- **IEEE 802.15:** only **physical and data-link** layers.

Piconet and scatternet The basic unit is a **piconet**: a **master** node and up to **seven active slave** nodes within about **10 m**. Multiple piconets can exist in the same large room and can be connected via a **bridge** node. An interconnected collection of piconets is a **scatternet**.

All communication is master-slave. Direct **slave-to-slave communication is not possible**.

Layered protocol architecture Protocols are grouped loosely into layers.

Layer / protocol	Role as taught
Radio (physical)	Matches OSI / 802 physical layer: radio transmission and modulation in ISM 2.4-2.485 GHz
Baseband	Analogous to the MAC sublayer , with some physical-layer elements. Master controls time slots and grouping into frames . Turns the raw bit stream into frames and defines key formats
Link Manager	Establishes logical channels: power management, authentication, QoS
L2CAP	Logical Link Control and Adaptation Protocol — shields upper layers from transmission details
Audio and control	Applications may reach them directly , without L2CAP
RFCOMM	Emulates a PC serial port (keyboard, mouse, modem, ...)
Telephony protocol	Real-time protocol for the three speech-oriented protocols ; manages call setup and termination
Service Discovery Protocol (SDP)	Locate services in the network
Applications and profiles	Top layer; use lower protocols

TDD slots. In the simplest form the master defines **625 μ s** slots. Master transmissions start in **even** slots, slave transmissions in **odd** slots — classical **TDM**, with the

master getting **half** the slots and slaves sharing the other half.

Each frame rides a logical **link** between master and slave. Two link types:

Link	Name	Use
ACL	Asynchronous Connection-Less	Packet-switched data at irregular intervals
SCO	Synchronous Connection-Oriented	Real-time data (e.g. telephone). Fixed slot in each direction. Frames are never retransmitted; forward error correction (FEC) provides reliability

L2CAP's three functions

1. Accept packets of up to **64 KB** from upper layers, **fragment** to frames, **reassemble** at the far end.
2. **Multiplex / demultiplex** multiple packet sources.
3. Handle **QoS** requirements.

Frame format A Bluetooth frame begins with an **access code** that usually **identifies the master**, so slaves in radio range of **two masters** can tell which traffic is theirs. Next is a **54-bit header** (typical MAC-sublayer fields), then a data field of up to **2,744 bits** for a **five-slot** transmission. For a **single time slot** the layout is the same except the data field is **240 bits**.

Header fields:

Field	Role
Address	Which of the eight active devices the frame is for
Type	Frame type: ACL, SCO, POLL, or NULL ; also error-correction type in the data field and how many slots the frame occupies
Flow Acknowledgement Sequence	Slave asserts when its buffer is full Piggybacks an ACK Numbers frames to detect retransmissions. Protocol is stop-and-wait , so one bit is enough
Header checksum	8-bit check

The entire 8-bit header is **repeated three times** to make the 54-bit header. The

receiver examines all three copies of each bit: if they agree, accept it; if not, **majority vote** wins.

SIG versions (all downward compatible) The SIG was formally announced on **20 May 1998**. Membership is given as **over 20,000 companies**. Founders: Ericsson, IBM, Intel, Toshiba, Nokia; many others joined later. **All versions support backward compatibility.**

Version	What the lecture stated
v1.0 and v1.0b	Basic Bluetooth; many problems ; manufacturers struggled to make products interoperable
v1.1	Ratified as IEEE 802.15.1-2002 . Fixed many 1.0b errors. Added non-encrypted channels and RSSI
v1.2	Faster connection and discovery . Adaptive Frequency Hopping (AFH) spread spectrum — skip crowded hop frequencies to resist RF interference. Practical speed up to 721 kbps (higher than v1.1). Ratified as IEEE 802.15.1-2005
v2.0 + EDR	Core spec 2004 . Enhanced Data Rate : nominal about 3 Mbps , practical 2.1 Mbps . Modulation: GFSK combined with PSK
v2.1 + EDR	Adopted 26 July 2007 . Headline feature: Secure Simple Pairing (SSP) — better pairing experience and stronger security
v3.0 + HS	Adopted 21 April 2009 . Theoretical transfer up to 24 Mbps , not over the Bluetooth link itself : Bluetooth is used for negotiation and establishment ; high-rate traffic rides an 802.11 link
v4.0	“Bluetooth Smart,” adopted 30 June 2010 . Includes Classic , Bluetooth High Speed , and Bluetooth Low Energy (LE) . LE, previously Wibree , is a subset of v4.0 with an entirely new protocol stack for rapid simple links
v4.1	Adopted 4 December 2013 . Software (not hardware) update to v4.0. Increased coexistence with LTE , bulk data-transfer rates, and devices in multiple roles simultaneously
v4.2	Released 2 December 2014 . Data length extension needs a hardware update; some older hardware can get features such as privacy updates via firmware

IEEE 802.15 WPAN standards IEEE **802.15** is the working group that specifies WPAN standards.

Standard	Topic as taught
802.15.1	WPAN Bluetooth — PHY and MAC for wireless connectivity with fixed, portable, and moving devices in personal operating space . Issued 2002 and 2005
802.15.2	Coexistence of WPANs with other unlicensed devices (e.g. WLANs). 802.15.2-2003 published 2003 ; Task Group 2 then went into hibernation
802.15.3	High-rate WPAN , MAC and PHY, 11-55 Mbps (2003). Amendments: 802.15.3a , 802.15.3b-2006 , 802.15.3c-2009
802.15.4	Low-rate WPAN (2003): low data rate, very long battery life , very low complexity . Defines OSI physical and data-link layers. Alternative PHY 802.15.4a-2007 ; revision 802.15.4-2006 (4b)
802.15.5	Mesh networking framework — interoperable, stable, scalable. Two parts: low-rate mesh on 802.15.4-2006 MAC ; high-rate mesh on 802.15.3b MAC
802.15.6	Body Area Network (BAN) . December 2011 draft. Devices in or around the human body : medical, consumer electronics, personal entertainment
802.15.7	Visible light communication
P802.15.8	Peer-aware communications
P802.15.9	Key management protocol
P802.15.10	Layer-2 routing

Security architecture Three **basic security services** in the Bluetooth standard:

Service	Meaning
Authentication	Verify identity of communicating devices from the Bluetooth device address . Bluetooth does not provide native user authentication
Confidentiality	Stop eavesdropping: only authorized devices access/view transmitted data
Authorization	A device must be authorized to use a service before it may do so

Bluetooth does **not** address **audit, integrity, or non-repudiation**. If needed, those come from **additional means**.

The **BR (Basic Rate) / EDR / HS** family defines **four security modes**. Each device must operate in one of them.

Security mode 1 — non-secure

- Authentication and encryption are **never initiated**.
- Devices are **indiscriminate**; nothing stops other Bluetooth devices from connecting.
- If a **remote** device **initiates** pairing, authentication, or encryption, a mode-1 device **will participate**.
- All **v2.0 and earlier** devices can support mode 1; **v2.1 and later** may use it only for **backward compatibility**.
- **NIST recommends never using security mode 1.**

Security mode 2 — service-level (after link, before logical channel)

- Security may start **after link establishment** but **before logical-channel establishment**.
- A local **security manager** (Bluetooth architecture) controls access to specific services.
- Centralized security manager holds **access-control policies** and interfaces to other protocols and users.

Security mode 3 — link-level enforced

- Security starts **before the physical link is fully established**.
- **Authentication and encryption are mandatory** for all connections to and from the device.
- All **v2.0 and earlier** can support mode 3; **v2.1 and later** only for **backward compatibility**.

Security mode 4 — service-level with SSP (v2.1 + EDR)

- Similar timing to mode 2: after **physical and logical link** setup.
- Uses **Secure Simple Pairing (SSP)**.
- **Elliptic-curve Diffie-Hellman (ECDH)** replaces **legacy key agreement** for **link-key generation**.
- Device **authentication and encryption algorithms remain those of v2.0 + EDR and earlier**.

Diagrams

Key terms

Term	Meaning in this lecture
WPAN	Wireless personal area network; Bluetooth's setting
SIG	Bluetooth Special Interest Group (1998; 1,500-page v1.0 in 1999)
Piconet	Master + up to 7 active slaves within ~10 m
Scatternet	Piconets joined by a bridge node
ACL / SCO	Async connectionless data vs synchronous connection-oriented real-time (no retransmission, FEC)
L2CAP	Segmentation, multiplexing, QoS
RFCOMM	Serial-port emulation
Access code	Identifies the master so overlapping piconets can be distinguished

Term	Meaning in this lecture
EDR	Enhanced Data Rate (v2.x): ~3 Mbps nominal, 2.1 Mbps practical, GFSK+PSK
HS	High Speed (v3.0): 24 Mbps over 802.11 after Bluetooth setup
LE / Wibree	Bluetooth Low Energy new stack inside v4.0 “Bluetooth Smart”
SSP	Secure Simple Pairing (v2.1); ECDH link keys in security mode 4
802.15.1-.10	IEEE WPAN family: Bluetooth, coexistence, high/low rate, mesh, BAN, VLC, PAC, KMP, L2 routing
Security modes 1-4	Non-secure; service-level; link-level mandatory crypto; SSP service-level

Lecture takeaways

- Bluetooth is a complete SIG stack on cheap short-range ISM radios; IEEE 802.15 only standardizes PHY/MAC.
- A piconet is strictly **star-shaped**: no slave-to-slave path except through the master (or a scatternet bridge).
- Baseband TDD (625 μ s), ACL vs SCO, and L2CAP fragmentation/mux/QoS are the operational core; audio/control can bypass L2CAP.
- Versions add hopping hygiene, EDR, SSP, 802.11 offload, LE, LTE coexistence, and longer PDUs — always backward compatible.
- Native security is **device** auth, confidentiality, and service authorization only. **Never use mode 1** (NIST). Mode 4 modernizes **key agreement** (ECDH) without changing the old auth/encryption algorithms.

M34: Virtual Private Network

Source: <https://www.youtube.com/watch?v=14DVrUgf2k8> **Instructor / expert:** Prof. Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

Learning objectives

- Contrast a leased-line private network with a VPN overlay on a public network.
- Name Access, Intranet, and Extranet VPNs and the four architectural components (client, NAS, VPN server, protocol).
- Reconstruct Layer-2 tunnel setup (PPP, LCP, PAP/CHAP, NCP) and the PPTP, L2F, and L2TP packet stories.
- Distinguish IPsec AH vs ESP and Layer-2 vs Layer-3 tunneling.
- State OpenVPN’s SSL/TLS properties and the lecture’s protocol preference; explain why firewalls and VPNs are used together.

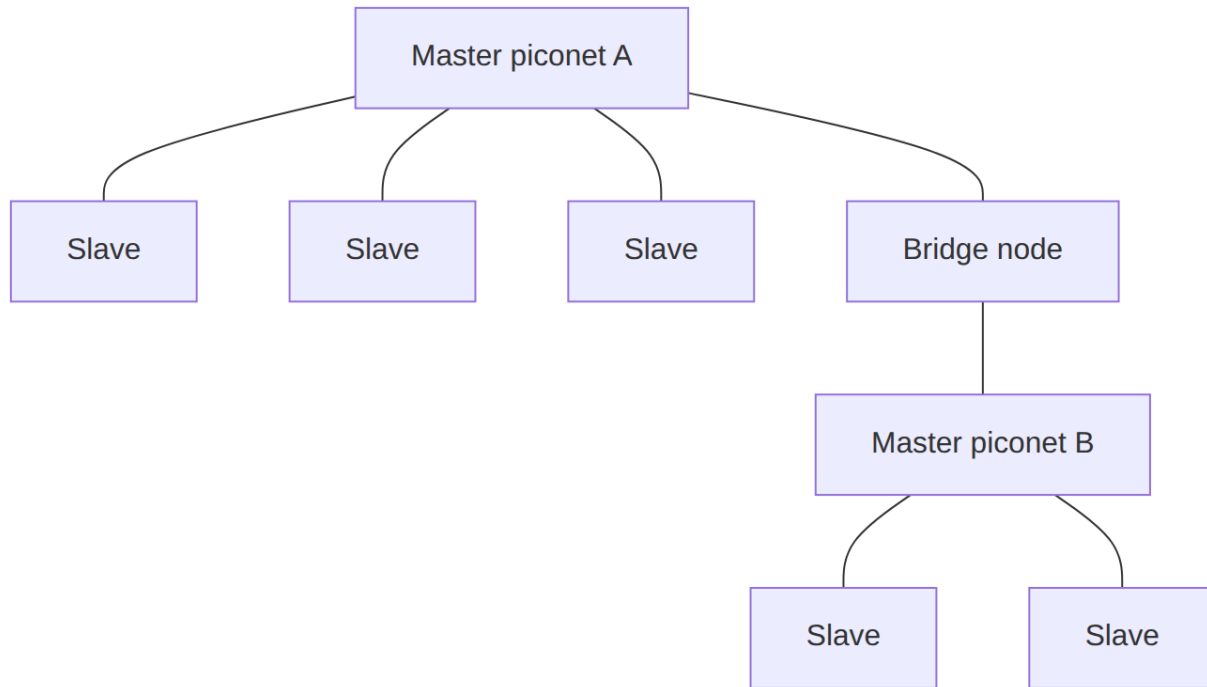


Figure 12: Architecture diagram

Core concepts

Almost all companies have offices and plants scattered over cities — sometimes countries. In the olden days it was common to **lease telephone lines** between locations. Some still do. A network built from **company computers and leased telephone lines** is a **private network**.

Private networks **work and are very secure**: traffic does not leak from company locations, and intruders must **physically wiretap** the lines. The problem is **cost**. When public data networks and then the **Internet** appeared, companies wanted to move data and voice onto the public network **without giving up private-network security**. That demand produced the **VPN**: an **overlay** on public networks with **most properties of private networks**.

A VPN provides a **secure connection** between sender and receiver over a **public, non-secure** network such as the Internet. They are **virtual** in the same sense as virtual circuits and virtual memory: an **illusion**, not a physical private plant.

A VPN is created by a **virtual point-to-point** path using **dedicated connections, virtual tunnels, and traffic encryption**. A VPN across the Internet is similar to a **WAN link** between sites. From the user's view, extended resources are used **the same way** as resources inside the private network.

Uses taught:

- Employees **securely access the company intranet** while traveling.
- Geographically separated offices join into **one cohesive network**.

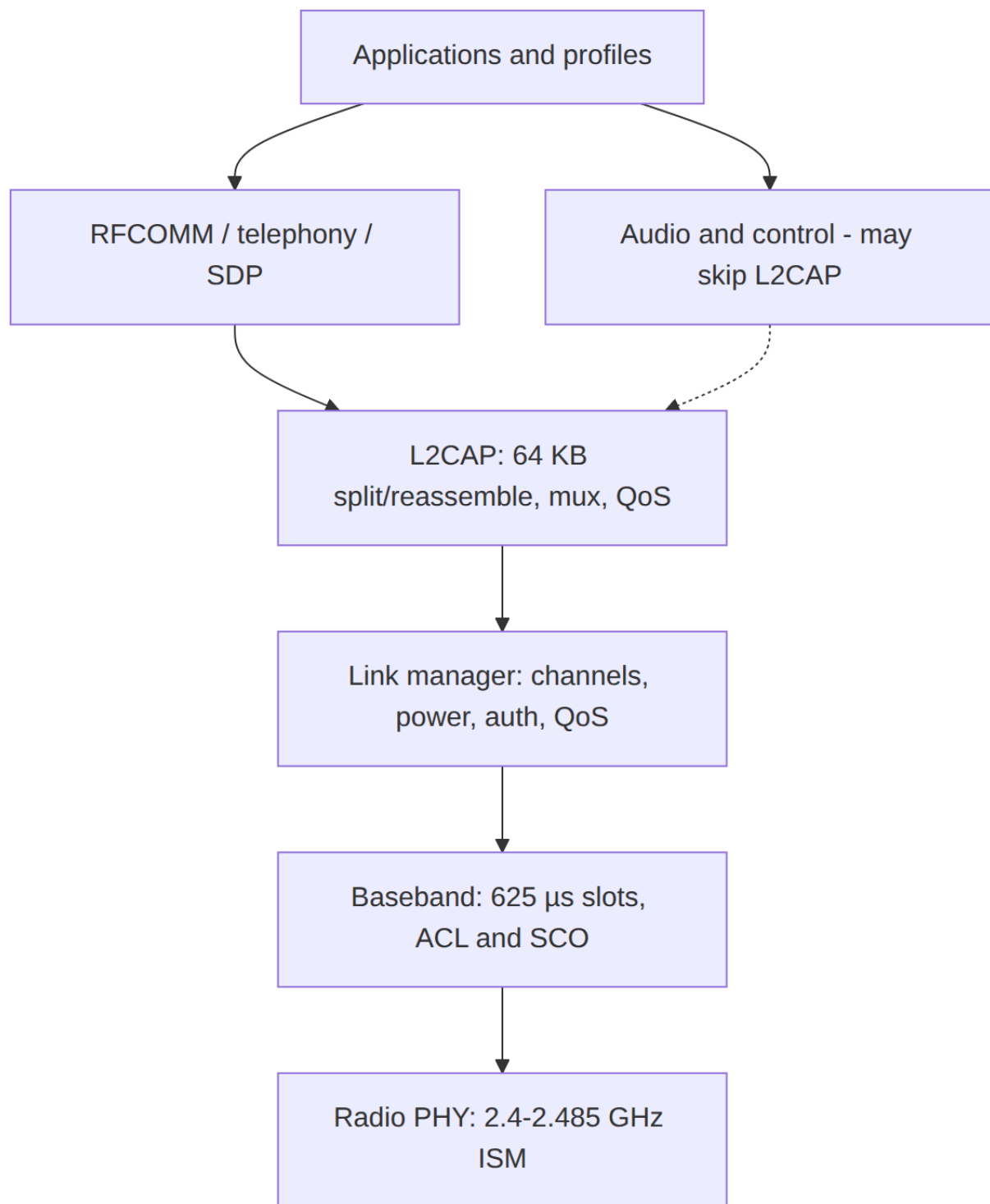


Figure 13: Architecture diagram

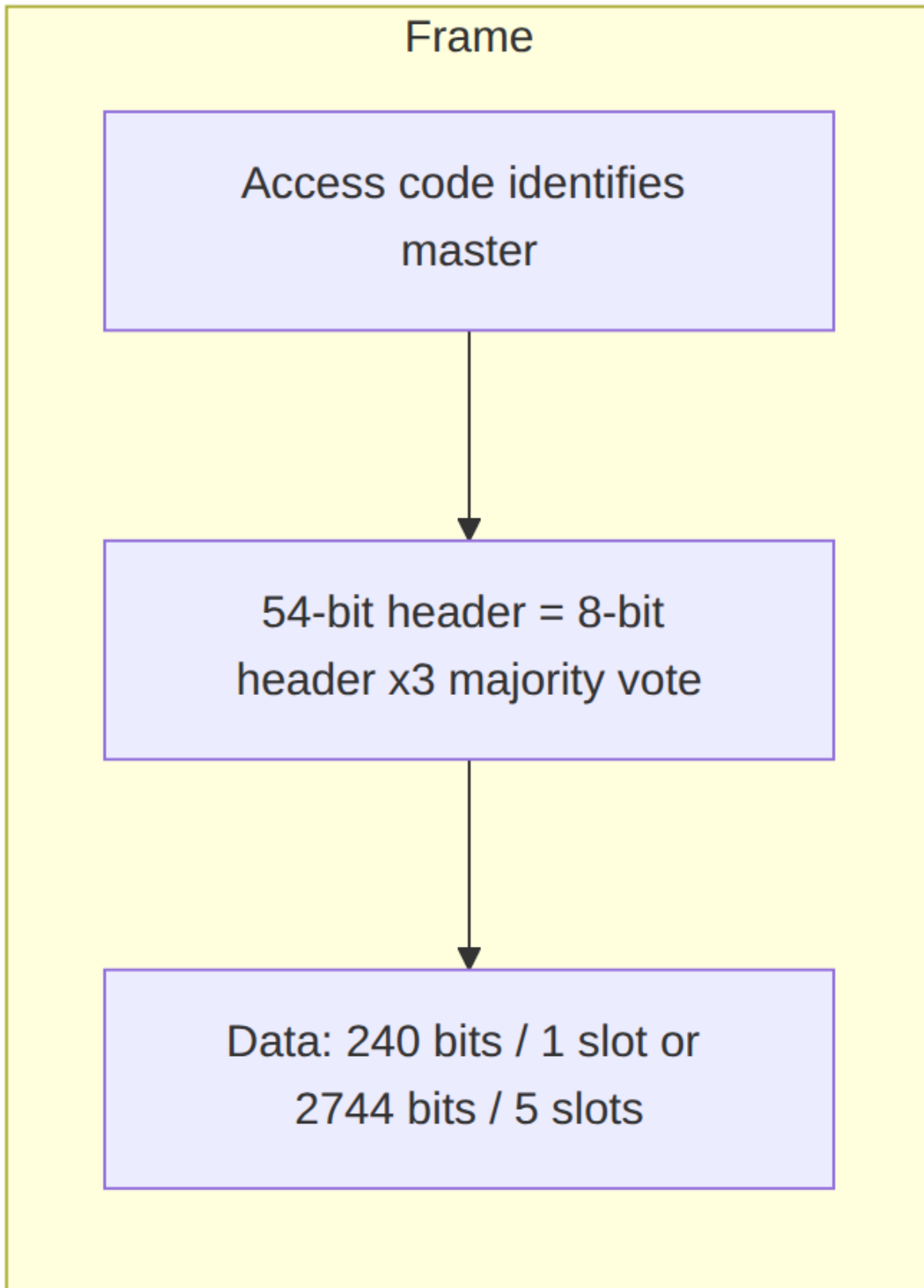


Figure 14: Architecture diagram

- Individuals secure **wireless transactions**, **circumvent geo-restrictions and censorship**, and connect to **proxy servers** to protect **personal identity and location**.

Three types of VPN All three use the Internet as a private enterprise backbone, but they serve different groups:

Type	Who it serves
Access VPN	Remote / mobile users, telecommuters, or a branch office needing reliable access to the corporate network
Intranet VPN	Branch offices linked to corporate headquarters securely
Extranet VPN	Customers, suppliers, and partners accessing the corporate intranet securely

Architecture (four components)

1. **VPN client**
2. **Network Access Server (NAS)**
3. **Tunnel-terminating device — VPN server**
4. **VPN protocol**

In a typical **Access VPN**:

- The client connects to the NAS over the **PSTN**.
- The remote user initiates a **PPP** connection with the ISP's NAS via PSTN.
- The NAS is owned by the ISP and usually sits in the ISP's **point of presence (PoP)**.
- After authentication, the NAS directs the packet into the **IP tunnel** between NAS and VPN server.
- The VPN server may sit in the **ISP PoP** or at the **corporate site**, depending on the model.
- The VPN server **recovers** the packet from the tunnel, **unwraps** it, and delivers it to the corporate network.

Two categories of tunneling protocols

Category	Protocols
Layer 2	PPTP, L2F, L2TP
Layer 3	IPsec

When **PPTP** and **L2F** were both submitted to the **IETF**, the features were **combined** into **L2TP**.

How Layer-2 tunneling works Layer-2 protocols operate at the **data-link layer**. They use security from **PPP**:

- Authentication: existing PPP methods **PAP** (Password Authentication Protocol) and **CHAP** (Challenge Handshake Authentication Protocol).
- **No specific provision for data encryption**; the user may encrypt **before** requesting VPN service.

General tunnel-creation process:

1. Client **dials** the NAS (typically in the ISP PoP) and starts PPP.
2. The NAS answers and performs **LCP** negotiation to establish an authenticated PPP connection.
3. Once PPP is up, the NAS uses **PAP or CHAP: PPP authentication request** → client **PPP authentication reply**.
4. If that succeeds, the NAS attempts to **open a PPP connection to the VPN server** (lecture: “open VPN request”).
5. The VPN server authenticates the NAS with **PAP or CHAP** (“open VPN reply”).
6. Client and VPN server then use **NCP** to negotiate the **network-layer** protocol.
7. Tunneling is complete: a VPN exists between client and corporate server.

PPTP (Point-to-Point Tunneling Protocol) Developed by **Microsoft** and equipment vendors including **Ascend Communications** and **3Com**, as an **extension of PPP**.

- Handles **only point-to-point** connections; **not** point-to-multipoint.
- NAS that lets the remote user start a VPN call is the **PPTP Access Concentrator (PAC)**.
- VPN server is the **PPTP Network Server (PNS)**.
- **No packet-by-packet encryption**; relies on **PPP’s native encryption** (taught in connection with PAP/CHAP).

PPTP packet (as drawn): PPP payload = data plus original **IP and TCP** headers. A PPTP packet (PPP payload + PPP header) is encapsulated in **GRE**, then an **IP header** is placed on top of GRE.

L2F (Layer 2 Forwarding) **Proprietary Cisco** protocol.

- **Protocol-independent**; can run over **X.25, Frame Relay, and ATM**.
- Supports **private IP, IPX, and AppleTalk**; uses **UDP** for Internet tunneling.
- **Many connections inside one tunnel**.
- VPN server is called the **Home Gateway**; other pieces match the general Layer-2 picture.
- Uses **PPP for dial-up user authentication**.
- Unlike PPTP, L2F defines **its own encapsulation header, not dependent on IP and GRE**, so it can work on different network types.
- Packet: **SLIP or PPP payload** in an L2F packet with **L2F header** and optional **L2F checksum trailer**.

L2TP (Layer 2 Tunneling Protocol) Combines PPTP and L2F.

- Runs over **UDP**, not GRE — many **firewalls do not support GRE**, so L2TP is **more firewall-friendly** than PPTP.
- NAS = **L2TP Access Concentrator (LAC)**; VPN server = **L2TP Network Server (LNS)**.
- Uses PPP dial-up and **PAP and CHAP**; also allows **RADIUS**.
- Permits **multiple tunnels** between the **same pair of endpoints**, each with a different **QoS**.
- Packet (as taught): **PPP header**, then **UDP**, then **IP**.
- Supported by many vendors; expected to be the **predominant** protocol once standardized.
- Relies on **IPsec** for data encryption. If the remote end **does not support IPsec**, it falls back to **less secure PPP encryption**.

Layer 3: IPsec IPsec was designed to **add security to TCP/IP**. It provides **packet-level authentication, integrity, and confidentiality** via two headers:

Header	Provides	Notes
AH (Authentication Header)	Header integrity and authentication, without confidentiality	No data encryption. Useful when only authentication is required. Authentication is not export-regulated , so AH is the preferred VPN method in those environments. Lower processing overhead than ESP
ESP (Encapsulating Security Payload)	Integrity, authentication, and confidentiality of the payload	Packet-by-packet encryption and a standards-based key-management protocol. Used when data encryption is desired

An IPsec VPN can be created with **AH, ESP, or both**.

Layer 2 versus Layer 3

Layer 2 (PPTP, L2F, L2TP)	Layer 3 (IPsec)
Traffic can carry non-IP (IPX, AppleTalk)	IP only

Layer 2 (PPTP, L2F, L2TP)	Layer 3 (IPsec)
Dial-up Individual dial-up access via PPP user authentication — seamless through ISPs	Designed for security between routers and firewalls; does not provide user authentication

OpenVPN (neither IPsec nor L2TP) A relatively new **open-source** product using the **OpenSSL** library and **SSL v3 or TLS v1**, plus an amalgam of other technologies.

- Supports **neither** Layer-2 nor Layer-3 tunneling protocols in the IPsec/L2TP/PPTP sense.
- **Not compatible** with IPsec, L2TP, or PPTP.
- **Highly configurable**. Runs best on **UDP**, but can use **any port**, including **TCP 443**.
- OpenSSL ciphers taught: **AES, Blowfish, 3DES, CAST-128, Camellia**, and more.

Capabilities listed:

- Tunnel any **IP subnet** or **virtual Ethernet adapter** over a **single UDP or TCP port**.
- Configure a **scalable, load-balanced** VPN server on one or more machines for **thousands of dynamic** client connections.
- Use all OpenSSL **encryption, authentication, and certification** features to protect private traffic on the Internet.
- Any cipher, key size, or **HMAC digest** OpenSSL supports for datagram integrity.
- **Static-key** conventional encryption **or certificate-based public-key** encryption.
- Static **pre-shared keys** or **TLS-based dynamic key exchange**.
- **Real-time adaptive link compression** and **traffic shaping**.
- Tunnel networks whose public endpoints are **dynamic** (DHCP or dial-in clients).
- Tunnel through **connection-oriented stateful firewalls** without explicit extra rules; tunnel over **NAT**.
- **Secure Ethernet bridges** using virtual **TAP** devices.
- Control via a **GUI on Windows or Mac**.

Compared with PPTP and L2TP/IPsec, generic OpenVPN can be **fiddly to set up**: install the client **and** extra configuration files. Many VPN providers ship **customized clients** to avoid that.

Which protocol to choose (instructor's ranking)

Protocol	Verdict in this lecture
PPTP	Very insecure — avoid , despite easy setup and cross-platform reach

Protocol	Verdict in this lecture
L2TP/IPsec	Same convenience, much more secure . Good for non-critical use and a quick setup without extra software , especially mobile devices where OpenVPN support is “somewhat patchy”
OpenVPN	Best all-round : reliable, fast, and most importantly secure , despite third-party software and more settings

Wherever possible, always choose OpenVPN. For a quick shield (phone vs casual criminals, public Wi-Fi), L2TP/IPsec will “probably do,” but given OpenVPN apps — **especially Android** — still **prefer OpenVPN**.

Firewalls and VPNs together Firewalls and VPNs **go hand in hand**.

- VPNs open **secure tunnels**.
- Firewalls sit at **strategic locations** and **block certain traffic**. Many firewall products provide **encrypted firewall-to-firewall tunnels**.
- Firewalls **control access** to corporate resources and **establish trust** between user and network.

Two sites each with a firewall still send data **vulnerable on the Internet**. A VPN supplies **privacy** between sites when there is **usually no trust** between the two sides. **Firewall + VPN** establishes **trust and privacy** and is **more secure than either alone**.

Older firewalls offered only firewall service; **many new firewall products support VPN functionality**. Both are needed for **effective security control**.

Diagrams

Key terms

Term	Meaning in this lecture
Private network	Company hosts plus leased telephone lines
VPN	Overlay tunnels + encryption on a public non-secure network
NAS	Network Access Server at the ISP PoP
PAC / PNS	PPTP Access Concentrator / PPTP Network Server
LAC / LNS	L2TP Access Concentrator / L2TP Network Server
Home Gateway	L2F name for the VPN server
PAP / CHAP	PPP authentication used by Layer-2 VPNs
LCP / NCP	PPP link control / network control (IPCP etc.)
GRE	PPTP encapsulation; many firewalls lack it
AH / ESP	IPsec authentication-only vs encrypting payload
OpenVPN	SSL/TLS VPN via OpenSSL; not interoperable with IPsec/L2TP/PPTP
TAP	Virtual Ethernet device for OpenVPN bridging

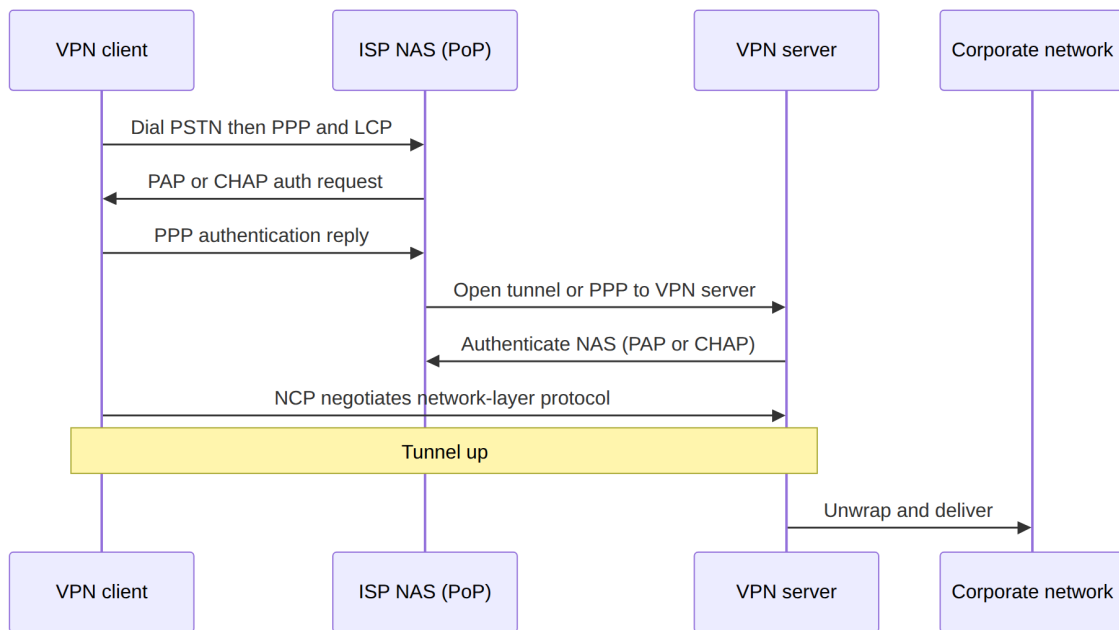


Figure 15: Architecture diagram

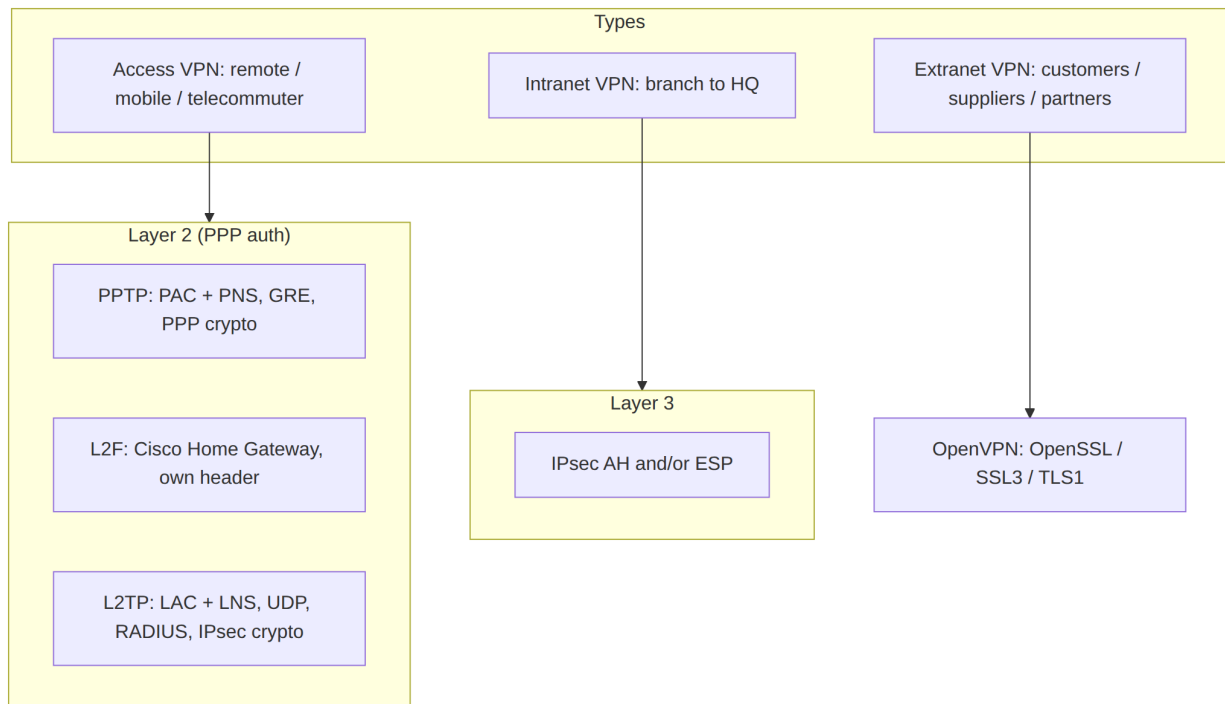


Figure 16: Architecture diagram

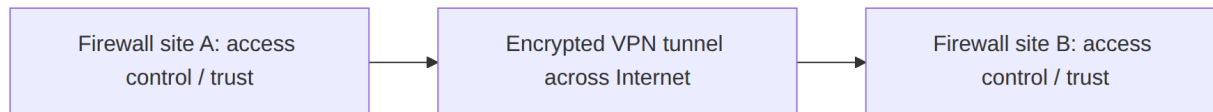


Figure 17: Architecture diagram

Lecture takeaways

- VPNs buy private-network **properties** without leased-line **cost**, via tunnels and encryption on the Internet.
- Layer 2 (PPTP/L2F/L2TP) is the dial-up/PPP world and can carry non-IP; IPsec is IP-only, router/firewall oriented, and has **no user authentication**.
- IETF folded PPTP+L2F into **L2TP**; encrypt L2TP with **IPsec** when possible.
- **Avoid PPTP**. Prefer **OpenVPN**; use **L2TP/IPsec** only for quick, non-critical, especially mobile, setups.
- A firewall pair without a tunnel still leaks on the Internet; **firewall + VPN** is stronger than either mechanism alone.

M35: WiMAX Technology and its Security

Source: https://www.youtube.com/watch?v=K8I_4d-oGhA **Instructor / expert:** Prof. Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

Playlist metadata names this module **WiMAX Technology and its Security**. The auto-generated captions attached to the video (and thus this transcript) are a near-duplicate of **Module 33**: they open with Bluetooth WPAN and never mention WiMAX or IEEE 802.16. Notes below follow **the spoken captions**, with the same ASR corrections as M33. They do not invent WiMAX material.

Learning objectives

- Place Bluetooth as a short-range UHF ISM WPAN technology and recount SIG / IEEE 802.15 origins.
- Draw a piconet and scatternet and state the master/slave communication rule.
- Walk the Bluetooth protocol stack, ACL vs SCO links, L2CAP roles, and the 54-bit header frame.
- Recite SIG versions from 1.0 through 4.2 and the IEEE 802.15.x WPAN family.
- Name Bluetooth's three native security services and the four BR/EDR/HS security modes, including the recommendation never to use mode 1.

Core concepts

Bluetooth is a wireless standard for exchanging data over **short distances** using **short-wavelength UHF** radio. It operates in the **ISM band from 2.4 to 2.485**

GHz, for mobile devices and **personal area networks (PANs / WPANs)**.

Origins and naming In **1994**, **Ericsson** wanted to connect mobile devices to other devices **without cables**. With **IBM, Intel, Nokia, and Toshiba** it formed a **Special Interest Group (SIG)**. They developed a short-range, **low-power, inexpensive** radio standard for computing and communication devices and accessories.

The name **Bluetooth** is after the 10th-century king **Harald Bluetooth**, who united distant Danish tribes into one kingdom. The name was proposed in **1997** by **Jim Kardach**, who had developed a system allowing mobile phones to talk to computers.

The Bluetooth SIG defines manufacturing standards. In **July 1999** it issued a **1,500-page specification of version 1.0**. Shortly after, the IEEE WPAN working group **802.15** adopted that document as a basis.

- **Bluetooth SIG specification:** a **complete system**, physical layer through application layer.
- **IEEE 802.15:** only **physical and data-link** layers.

Piconet and scatternet The basic unit is a **piconet**: a **master** node and up to **seven active slave** nodes within about **10 m**. Multiple piconets can exist in the same large room and can be connected via a **bridge** node. An interconnected collection of piconets is a **scatternet**.

All communication is master-slave. Direct **slave-to-slave communication is not possible**.

Layered protocol architecture Protocols are grouped loosely into layers.

Layer / protocol	Role as taught
Radio (physical)	Matches OSI / 802 physical layer: radio transmission and modulation in ISM 2.4-2.485 GHz
Baseband	Analogous to the MAC sublayer , with some physical-layer elements. Master controls time slots and grouping into frames . Turns the raw bit stream into frames and defines key formats
Link Manager	Establishes logical channels: power management, authentication, QoS
L2CAP	Logical Link Control and Adaptation Protocol — shields upper layers from transmission details
Audio and control	Applications may reach them directly , without L2CAP
RFCOMM	Emulates a PC serial port (keyboard, mouse, modem, ...)

Layer / protocol	Role as taught
Telephony protocol	Real-time protocol for the three speech-oriented protocols ; manages call setup and termination
Service Discovery Protocol (SDP) Applications and profiles	Locate services in the network Top layer; use lower protocols

TDD slots. In the simplest form the master defines **625 µs** slots. Master transmissions start in **even** slots, slave transmissions in **odd** slots — classical **TDM**, with the master getting **half** the slots and slaves sharing the other half.

Each frame rides a logical **link** between master and slave. Two link types:

Link	Name	Use
ACL	Asynchronous Connection-Less	Packet-switched data at irregular intervals
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L2CAP's three functions

1. Accept packets of up to **64 KB** from upper layers, **fragment** to frames, **reassemble** at the far end.
2. **Multiplex / demultiplex** multiple packet sources.
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Header fields:

Field	Role
Address	Which of the eight active devices the frame is for

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Acknowledgement	Piggybacks an ACK
Sequence	Numbers frames to detect retransmissions. Protocol is stop-and-wait , so one bit is enough
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Standard	Topic as taught
802.15.1	WPAN Bluetooth — PHY and MAC for wireless connectivity with fixed, portable, and moving devices in personal operating space . Issued 2002 and 2005
802.15.2	Coexistence of WPANs with other unlicensed devices (e.g. WLANs). 802.15.2-2003 published 2003 ; Task Group 2 then went into hibernation
802.15.3	High-rate WPAN , MAC and PHY, 11-55 Mbps (2003). Amendments: 802.15.3a , 802.15.3b-2006 , 802.15.3c-2009
802.15.4	Low-rate WPAN (2003): low data rate, very long battery life , very low complexity . Defines OSI physical and data-link layers. Alternative PHY 802.15.4a-2007 ; revision 802.15.4-2006 (4b)
802.15.5	Mesh networking framework — interoperable, stable, scalable. Two parts: low-rate mesh on 802.15.4-2006 MAC ; high-rate mesh on 802.15.3b MAC
802.15.6	Body Area Network (BAN) . December 2011 draft (captions once say “802.15.4 task group”). Devices in or around the human body : medical, consumer electronics, personal entertainment
802.15.7	Visible light communication
P802.15.8	Peer-aware communications
P802.15.9	Key management protocol
P802.15.10	Layer-2 routing

Security architecture Three **basic security services** in the Bluetooth standard:

Service	Meaning
Authentication	Verify identity of communicating devices from the Bluetooth device address . Bluetooth does not provide native user authentication
Confidentiality	Stop eavesdropping: only authorized devices access/view transmitted data
Authorization	A device must be authorized to use a service before it may do so

Bluetooth does **not** address **audit, integrity, or non-repudiation**. If needed, those come from **additional means**.

The **BR (Basic Rate) / EDR / HS** family defines **four security modes**. Each device must operate in one of them.

Security mode 1 — non-secure

- Authentication and encryption are **never initiated**.
- Devices are **indiscriminate**; nothing stops other Bluetooth devices from connecting.
- If a **remote** device **initiates** pairing, authentication, or encryption, a mode-1 device **will participate**.
- All **v2.0 and earlier** devices can support mode 1; **v2.1 and later** may use it only for **backward compatibility**.
- The lecture's recommendation (NIST in the cleaner caption pass): **never use security mode 1**.

Security mode 2 — service-level (after link, before logical channel)

- Security may start **after link establishment** but **before logical-channel establishment**.
- A local **security manager** (Bluetooth architecture) controls access to specific services.
- Centralized security manager holds **access-control policies** and interfaces to other protocols and users.

Security mode 3 — link-level enforced

- Security starts **before the physical link is fully established**.
- **Authentication and encryption are mandatory** for all connections to and from the device.
- All **v2.0 and earlier** can support mode 3; **v2.1 and later** only for **backward compatibility**.

Security mode 4 — service-level with SSP (v2.1 + EDR)

- Similar timing to mode 2: after **physical and logical link** setup.
- Uses **Secure Simple Pairing (SSP)**.
- **Elliptic-curve Diffie-Hellman (ECDH)** replaces **legacy key agreement** for **link-key generation**.

- Device **authentication and encryption algorithms remain those of v2.0 + EDR and earlier.**

Diagrams

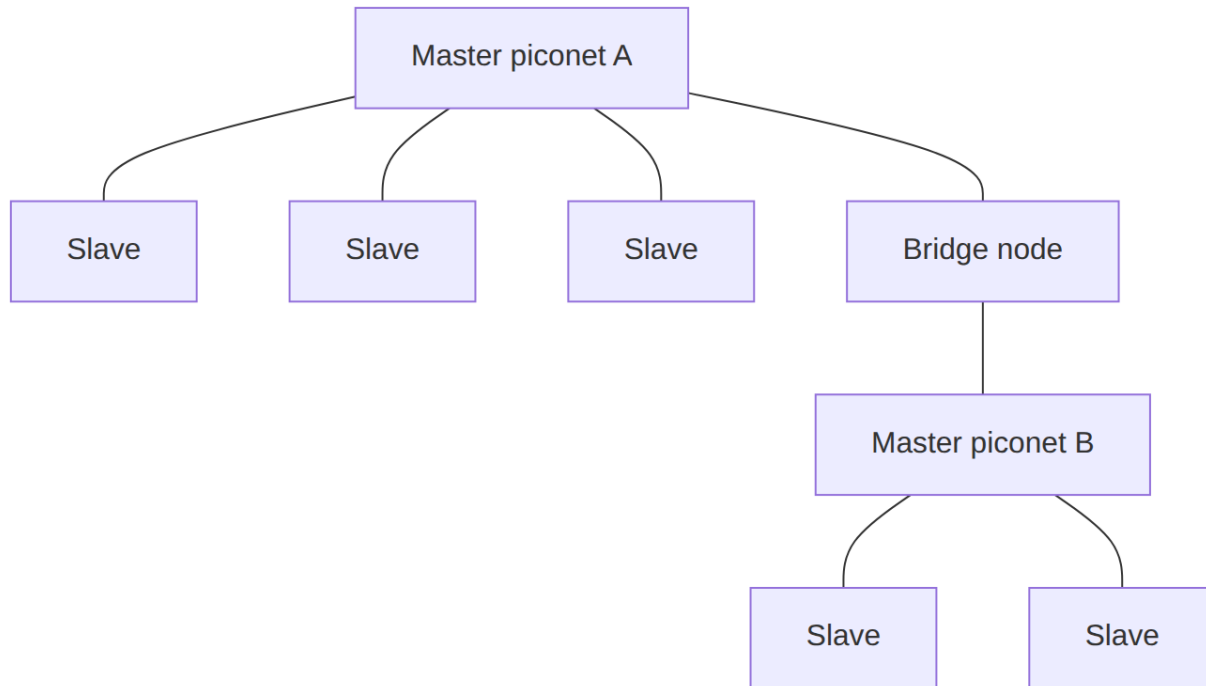


Figure 18: Architecture diagram

Key terms

Term	Meaning in this lecture
WPAN	Wireless personal area network; Bluetooth's setting
SIG	Bluetooth Special Interest Group (1998; 1,500-page v1.0 in 1999)
Piconet	Master + up to 7 active slaves within ~10 m
Scatternet	Piconets joined by a bridge node
ACL / SCO	Async connectionless data vs synchronous connection-oriented real-time (no retransmission, FEC)
L2CAP	Segmentation, multiplexing, QoS
RFCOMM	Serial-port emulation
Access code	Identifies the master so overlapping piconets can be distinguished
EDR	Enhanced Data Rate (v2.x): ~3 Mbps nominal, 2.1 Mbps practical, GFSK+PSK
HS	High Speed (v3.0): 24 Mbps over 802.11 after Bluetooth setup
LE / Wibree	Bluetooth Low Energy new stack inside v4.0 "Bluetooth Smart"
SSP	Secure Simple Pairing (v2.1); ECDH link keys in security mode 4

Term	Meaning in this lecture
802.15.1-.10	IEEE WPAN family: Bluetooth, coexistence, high/low rate, mesh, BAN, VLC, PAC, KMP, L2 routing
Security modes 1-4	Non-secure; service-level; link-level mandatory crypto; SSP service-level

Lecture takeaways

- Captions for this playlist slot teach **Bluetooth**, not WiMAX: complete SIG stack on cheap short-range ISM radios; IEEE 802.15 only standardizes PHY/MAC.
- A piconet is strictly **star-shaped**: no slave-to-slave path except through the master (or a scatternet bridge).
- Baseband TDD (625 μ s), ACL vs SCO, and L2CAP fragmentation/mux/QoS are the operational core; audio/control can bypass L2CAP.
- Versions add hopping hygiene, EDR, SSP, 802.11 offload, LE, LTE coexistence, and longer PDUs — always backward compatible.
- Native security is **device** auth, confidentiality, and service authorization only. **Never use mode 1**. Mode 4 modernizes **key agreement** (ECDH) without changing the old auth/encryption algorithms.

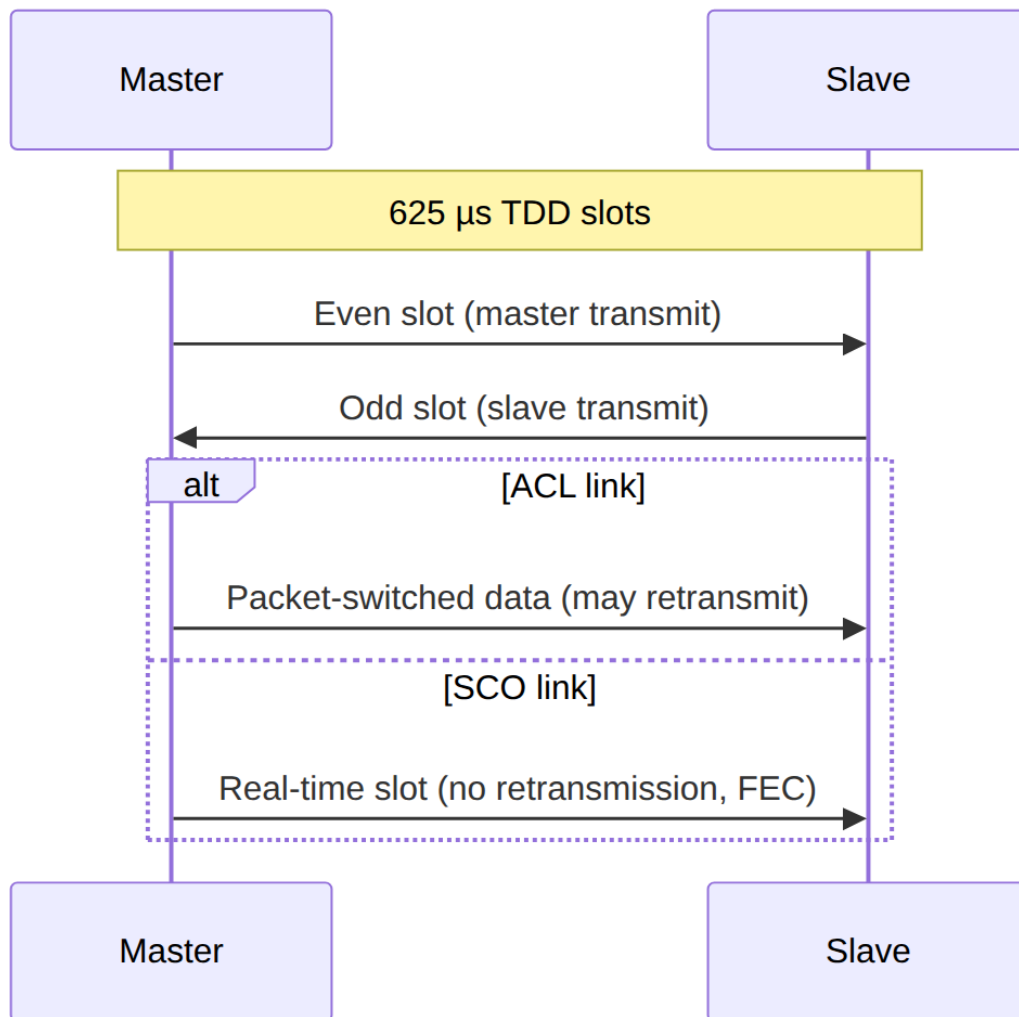


Figure 19: Architecture diagram

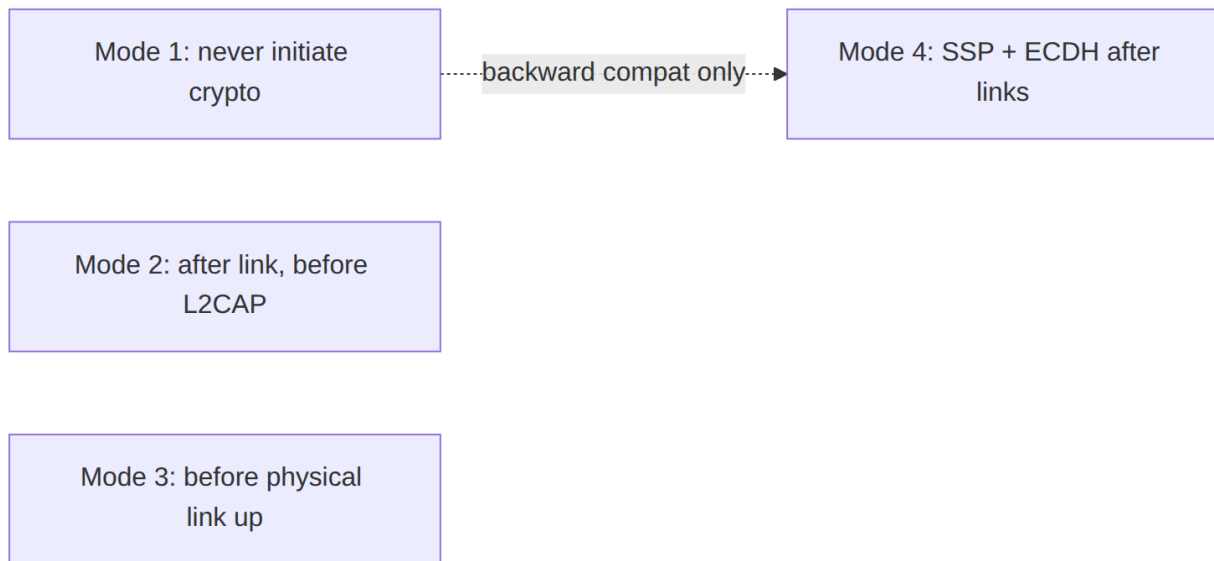


Figure 20: Architecture diagram